

ANSI/AGMA 6135-A02

Metric Edition

AMERICAN NATIONAL STANDARD

*Design, Rating and Application of
Industrial Globoidal Wormgearing (Metric
Edition)*

ANSI/AGMA 6135-A02



AGMA STANDARD

American National Standard

Design, Rating and Application of Industrial Globoidal Wormgearing (Metric Edition)

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Approved March 20, 2003

ABSTRACT

This standard provides guidelines for the design, rating and application of globoidal wormgearing mounted with axes at a 90 degree angle. Specific definitions for globoidal wormgearing terms are presented, along with formulas for determining the geometric sizes of the major features for the worm and gear. Design considerations, design procedures, gear blanks and self-locking conditions are also discussed. Procedures for rating the load capacity of globoidal wormgearing are included.

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Foreword

[The foreword, footnotes and annexes, if any, in this document are provided for informational purposes only and are not to be construed as a part of ANSI/AGMA Standard 6135-A02, *Design, Rating and Application of Industrial Globoidal Wormgearing (Metric Edition)*.]

This standard revises, combines and supersedes two previous independent standards: ANSI/AGMA 6030-C87, *Design of Industrial Double-Enveloping Wormgears*, and ANSI/AGMA 6017-E86, *Rating and Application of Single and Multiple Reduction Double-Enveloping Worm and Helical-Worm Speed Reducers*. It presents guidelines and practices for the design, rating and application for industrial globoidal wormgearing.

This Standard has been revised to conform with the ANSI (American National Standard Institute) style manual, to add metric dimensions and expand the definitions section.

The first draft of ANSI/AGMA 6135-A02 was made in December, 2000. It was approved by the AGMA membership in October 2001. It was approved as an American National Standard on March 20, 2003.

Suggestions for improvement of this standard will be welcome. They should be sent to the American Gear Manufacturers Association, 500 Montgomery Street, Suite 350, Alexandria, Virginia 22314.

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American National Standard – Design, Rating and Application of Industrial Globoidal Wormgearing (Metric Edition)

1 Scope

This standard provides guidelines for the design, rating and application of globoidal wormgearing mounted with axes at a 90 degree angle. This type of gearing is called double enveloping wormgearing in the United States, but the term globoidal is more commonly used worldwide.

This standard discusses the two main categories of globoidal wormgearing manufactured in industry today: the traditional Hindley designs, and the more recently developed full contact designs. The standard presents the formulas for calculating general gearset proportions for the globoidal designs and a rating procedure for Hindley designs. It does not provide a detailed or complete procedure for the design of gearsets and the cutting tools required.

Globoidal wormgearing of the Hindley design has been used in industrial applications for over 200 years. The manufacturers of Hindley wormgearing have each designed and built their own tooling to manufacture the gearing. The Hindley design uses a common base circle to form all of the tooth profiles. The worm thread is generated by a cutting edge (line) rotating in a single plane.

The more recently developed full contact designs do not use a constant base circle to form the tooth profiles. The worm thread is generated in multiple planes by a generating surface. The manufacturers of full contact gearing have each conducted their own research, and the details of their designs are proprietary to each company. This design may

provide increased load carrying capability over the Hindley design, but a standard rating method for full contact designs has not been established.

1.1 Types of reducers

This standard provides information pertaining to globoidal wormgear speed reducers of the following specific types:

- single reduction reducers incorporating globoidal wormgearing;
- multiple reduction reducers incorporating globoidal wormgearing in each stage of reduction;
- the wormgear portion of multiple reduction reducers incorporating globoidal wormgearing with other types of gearing.

1.2 Speed limitation

The velocity factor table in this standard provides data for sliding velocities up to 30 meters per second. For velocities in excess of 30 meters per second, the manufacturer should be consulted.

Where the sliding velocity of the worm thread exceeds 10 meters per second, additional consideration must be given to the lubrication of gears and bearings. Special lubricants and a pressurized lubrication system may be required.

1.3 Included ratios

The input power formulas for globoidal wormgearing apply for ratios of 3:1 and greater. Engineering practice dictates that ratios in excess of 74:1 should be discussed with the manufacturer.

1.4 Exceptions

Wildhaber plane enveloping globoidal gearing is not covered in this standard.

This standard does not pertain to globoidal wormgearing used as speed increasers. For such applications, the manufacturer should be consulted.

1.5 Intended use

The equations and values presented provide a general approach to design and rating. Deviations from the methods and values stated in this manual may be made when justified by experience, testing or more specific analysis. It is intended for use by experienced gear designers capable of selecting reasonable values based on their knowledge of the

performance of similar designs and the effect of such items as lubrication, deflection, manufacturing tolerances, metallurgy, residual stresses and system dynamics. It is not intended for use by the engineering public at large.

1.6 References

The following standards contain provisions which, through reference in this text, constitute provisions of this American National Standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this American National Standard are encouraged to investigate the possibility of applying the most recent editions of the standards indicated below.

ANSI/AGMA 2004-B89, *Gear Materials and Heat Treatment Manual*

ANSI/AGMA 6001-D97, *Design and Selection of Components for Enclosed Gear Drives*

ANSI/AGMA 6110-F97, *Standard for Spur, Helical, Herringbone and Bevel Enclosed Drives*

ANSI/AGMA 9005-D94, *Industrial Gear Lubrication*

2 Symbols, terms and definitions

2.1 Symbols

The symbols, terms and definitions, when applicable, conform to ANSI/AGMA 1012-F90, *Gear Nomenclature, Definitions Of Terms With Symbols*. See table 1.

NOTE: The symbols and terminology used in this standard may differ from other AGMA standards. The user should not assume that familiar symbols can be used without careful study of table 1.

Table 1 - Symbols used in design of industrial globoidal wormgears

ISO Symbols	Description	Units	First used	Ref. para.
a	Center distance	mm	Eq. 2	2.2
b_{f1}	Length of flat on outside diameter of worm	mm	Eq. 22	2.2
b_1	Worm face width	mm	- -	2.2
b_2	Gear face width	mm	- -	2.2
b_{1eff}	Effective worm thread length	mm	Eq. 21	2.2
c_1	Clearance in the worm	mm	Eq. 13	2.2
c_2	Clearance in the gear	mm	Eq. 14	2.2
d_{a1}	Worm throat diameter in transverse plane	mm	Eq. 17	2.2
d_{a2}	Gear throat diameter in central plane	mm	- -	2.2
d_b	Base diameter (Hindley design)	mm	Eq. 20	2.2
d_{e1}	Worm outside diameter	mm	- -	2.2
d_{e2}	Gear outside diameter	mm	- -	2.2
d_{f1}	Worm root diameter in transverse plane	mm	Eq. 3	2.2
d_{f2}	Gear root diameter in central plane	mm	- -	2.2
d_{w1}	Worm pitch diameter in transverse plane	mm	Eq. 2	2.2
d_{w2}	Gear pitch diameter in central plane	mm	Eq. 4	2.2
h_{am1}	Addendum of worm thread	mm	Eq. 12	2.2
h_{am2}	Addendum of gear tooth	mm	Eq. 12	2.2
h_{fm1}	Dedendum of worm thread	mm	Eq. 15	2.2
h_{fm2}	Dedendum of gear tooth	mm	Eq. 16	2.2
h_w	Working depth of tooth	mm	Eq. 11	2.2
h_1	Whole depth of worm thread	mm	Eq. 10	2.2
h_2	Whole depth of gear tooth	mm	Eq. 10	2.2
j_x	Backlash	mm	- -	2.2
m_a	Axial module	mm	Eq. 6	4.7

(continued)

Table 1 (concluded)

ISO Symbols	Description	Units	First used	Ref. para.
m_n	Normal module	mm	Eq. 9	4.9
n_1	Rotational speed of worm	rpm	Eq. 27	- -
P_1	Input power rating	kW	Eq. 27	8.1.3
p_{n1}	Normal circular pitch	mm	Eq. 8	2.2
p_{t2}	Circular pitch	mm	Eq. 5	2.2
r_{f2}	Gear root form radius	mm	Eq. 24	2.2
r_{g1}	Worm throat form radius	mm	Eq. 25	2.2
r_{g2}	Gear throat form radius	mm	Eq. 26	2.2
s_{n2}	Normal gear tooth thickness	mm	- -	2.2
s_k	Minimum amount of gear bronze below tooth root	mm	- -	9.3
s_{mx1}	Axial worm thread thickness	mm	- -	2.2
s_{mx2}	Axial gear tooth thickness	mm	Eq. 20	2.2
s_{n1}	Normal worm thread thickness	mm	- -	2.2
u	Gear ratio	- -	Eq. 1	2.2
V_g	Sliding velocity	mps	Eq. 28	8.1.4
Z_m	Face width and material factor	- -	Eq. 27	8.1.3
Z_u	Ratio correction factor	- -	Eq. 27	8.1.3
Z_v	Velocity factor	- -	Eq. 27	8.1.3
Z_w	Basic pressure factor	- -	Eq. 27	8.1.3
z_2	Number of teeth in gear	- -	Eq. 1	4.1
z_1	Number of threads in worm	- -	Eq. 1	4.2
α_n	Normal pressure angle	degrees	Eq. 19	2.2
α_x	Axial pressure angle	degrees	Eq. 19	2.2
δ_{m1}	Lead angle at pitch point	degrees	Eq. 7	2.2

2.2 Terms and definitions

addendum, h_{am1} , h_{am2} : The addendum of the worm thread and gear tooth is the radial distance from the pitch circle to the addendum circle on the worm and gear, respectively. See figures 1, 2 and 4.

addendum circle, gear: The gear addendum circle coincides with the tops of the gear teeth and is defined in the central plane. See figures 1 and 2.

addendum circle, worm: The worm addendum circle coincides with the top of the worm thread and is defined in the transverse plane. See figure 4.

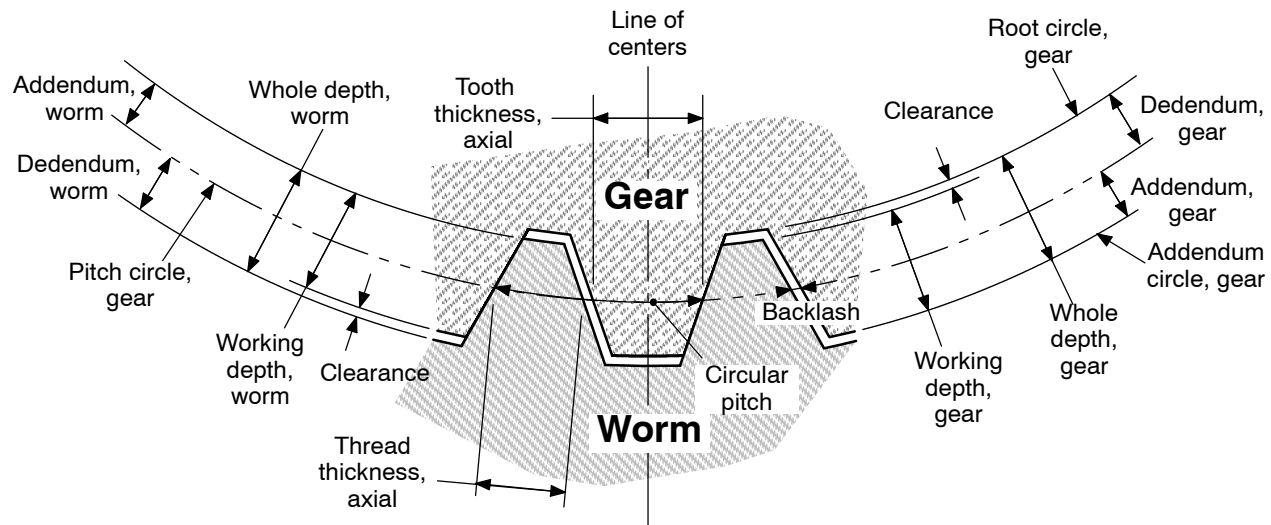


Figure 1 - Worm and gear mesh in central plane

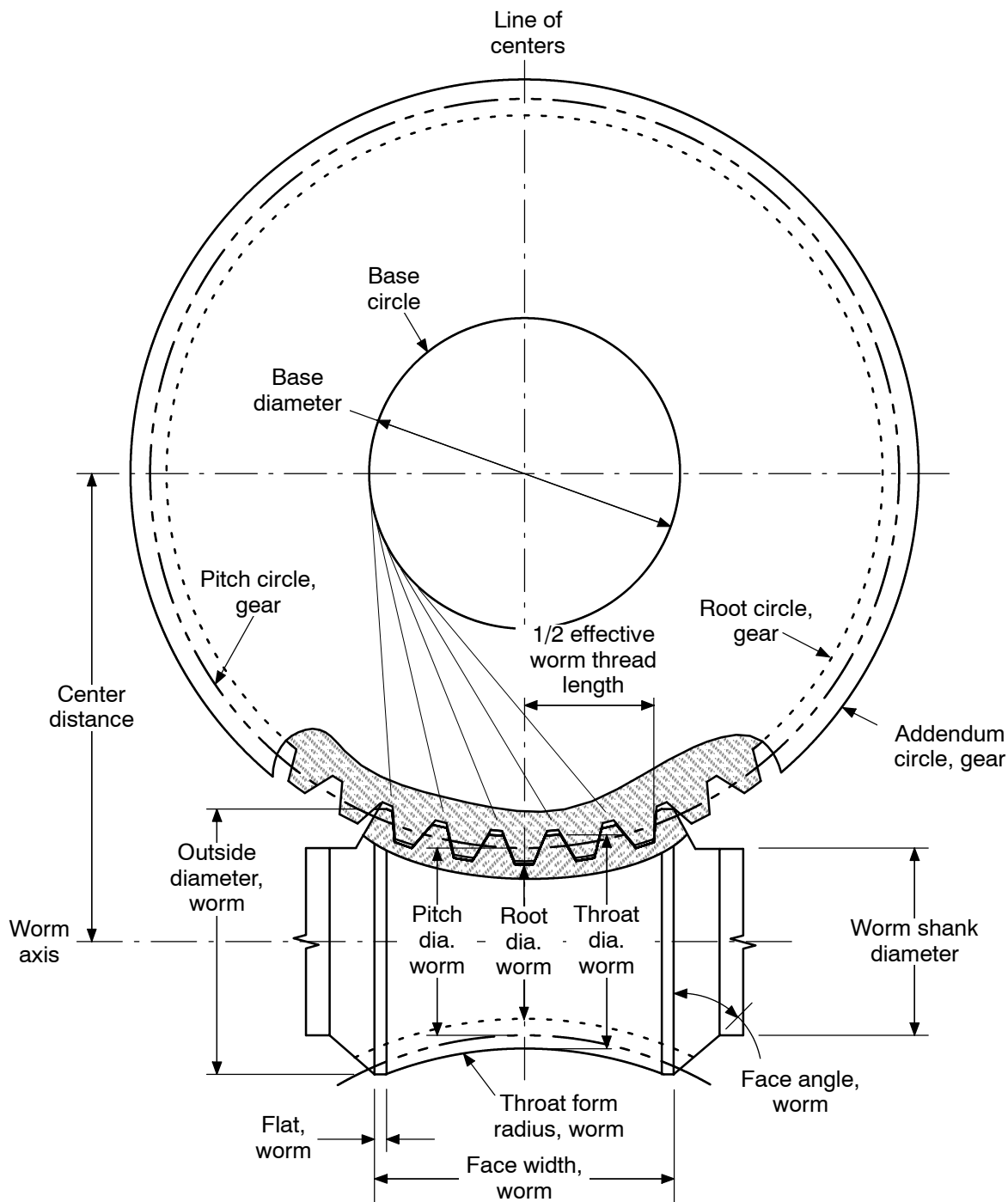


Figure 2 - Gearset central plane

backlash, j_x : The backlash is the amount by which the circular pitch exceeds the worm thread and gear tooth thickness. See figure 1 and annex C.

base circle: The base circle is the tangent circle of two generating lines of the tooth of the worm cutter. For the Hindley design, the generating lines are cutting lines which lie in the central plane. They are the cutting edges of the worm cutter. See figure 2.

For full contact designs, the generating lines are the generatrices of cutting planes or cutting cones.

base diameter, d_b : The base diameter is the diameter of the base circle. See figure 2.

center distance, a : The center distance of the gearset is the shortest distance from the axis of the worm to the axis of the gear. See figure 2.

central plane: The central plane is the plane perpendicular to the gear axis and contains the worm axis and the line of centers. See figures 1, 2 and 3.

circular pitch, p_{t2} : The circular pitch is the length of the arc along the gear pitch circle between two corresponding successive gear tooth flanks measured in the central plane. See figure 1.

circular pitch, normal, p_{n1} : The normal circular pitch is a theoretical value and is defined as the circular pitch times the cosine of the lead angle at the pitch point.

clearance, c_1, c_2 : The clearance is the amount by which the whole depth of the worm threads and gear teeth exceeds the working depth. The clearance is measured in the central plane. See figure 1.

dedendum, h_{fm1}, h_{fm2} : The dedendum of the worm thread and gear tooth is the radial distance from the root circle to the pitch circle on the worm and gear respectively. See figure 1.

effective worm thread length, b_{1eff} : The effective worm thread length is two times the distance from

the transverse plane to the point on the thread where theoretical contact with a gear tooth first occurs. It is measured parallel to the worm axis. See figure 2.

face angle, gear: The gear face angle is the angle of the beveled surface between the gear face and the gear outside diameter. See figure 4.

face angle, worm: The worm face angle is the angle of the beveled surface between the worm shank diameter and the worm outside diameter. See figure 2.

face width, gear, b_2 : The gear face width is the distance between the faces of the gear measured parallel to the gear axis. See figure 4.

face width, worm, b_1 : The worm face width is the distance across the worm flats measured parallel to the worm axis. See figure 2.

flats, gear: Gear flats are the surfaces included on the outside diameter of the gear. Gear flats avoid sharp edges at the intersections of the gear face angles and the throat form radius. See figure 4.

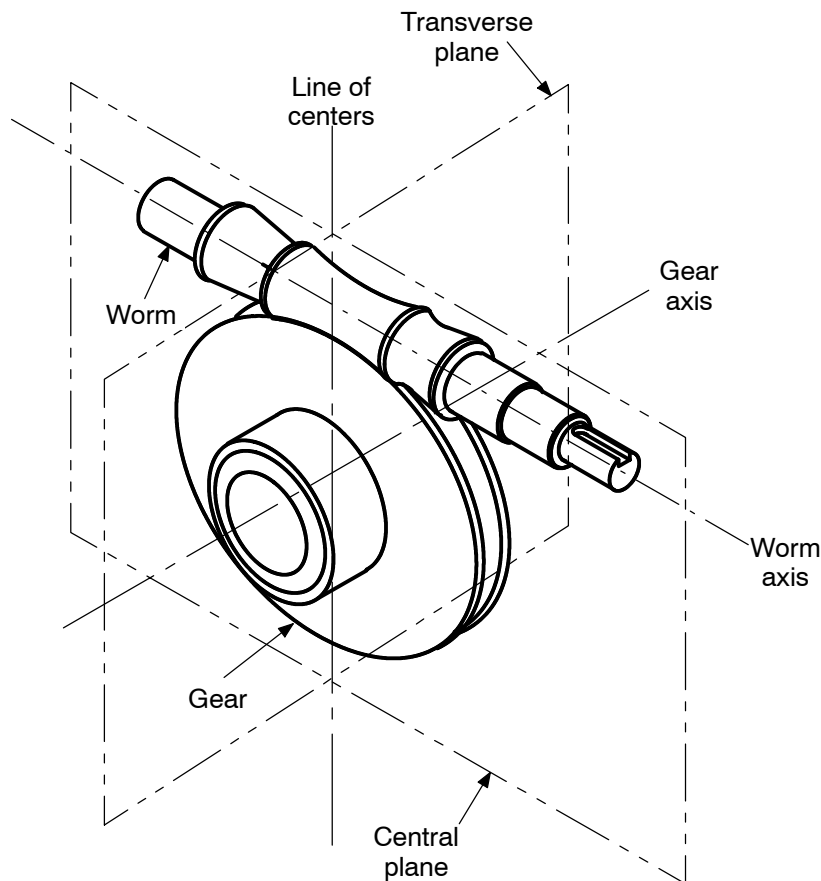


Figure 3 - Relationship of central plane and transverse plane

flats, worm, b_{f1} : Worm flats are the surfaces included on the outside diameter of the worm. Worm flats avoid sharp edges at the intersections of the worm face angles and the throat form radius. See figure 2.

gear: The gear is the toothed output member. The gear increases in diameter from the central plane toward both sides generally conforming to the curvature of the worm. Other commonly used terms to describe this component include wormwheel and wormgear. For purposes of this standard, the term gear is used. See figures 3 and 4.

gear axis: The gear axis is the axis of rotation of the

gear. The gear axis lies in the transverse plane. See figures 3 and 4.

gear face: The gear faces are the flat sides of the gear. See figure 4.

root form radius, gear, r_{f2} : The gear root form radius is the root radius generated by the hob. It is shown in the transverse plane and provides clearance for the tips of the worm threads. See figure 4.

lead angle at the pitch point, δ_{m1} : The lead angle at the pitch point is the angle formed between the tangent to the trace of the pitch point on the worm thread surface and a line perpendicular to the worm axis. The lead angle varies through the length and the depth of the worm. See figure 6.

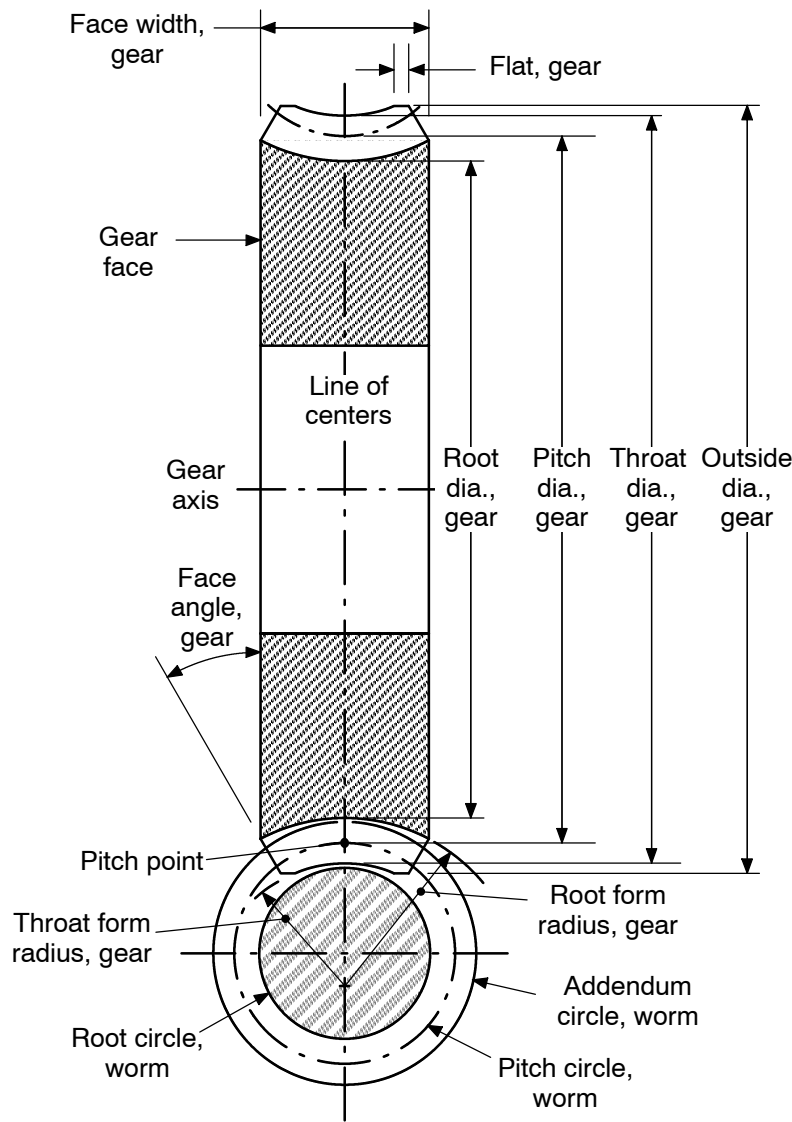


Figure 4 - Gearset transverse plane

line of centers: The line of centers connects the center of the worm pitch circle and the center of the gear pitch circle. The line of centers is formed by the intersection of the central plane and the transverse plane. See figure 3.

outside diameter, gear, d_{e2} : The gear outside diameter is the maximum diameter of the gear blank measured at the gear flats. See figure 4.

outside diameter, worm, d_{e1} : The worm outside diameter is the maximum diameter of the worm blank measured at the worm flats. See figure 2.

pitch circle, gear: The gear pitch circle is the reference circle used for determination of gear tooth proportions. The gear pitch circle lies in the central plane. See figures 1, 2 and 5.

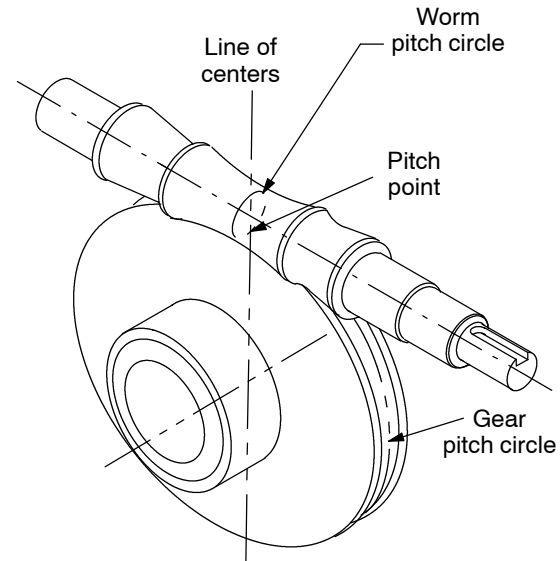


Figure 5 - Pitch circles

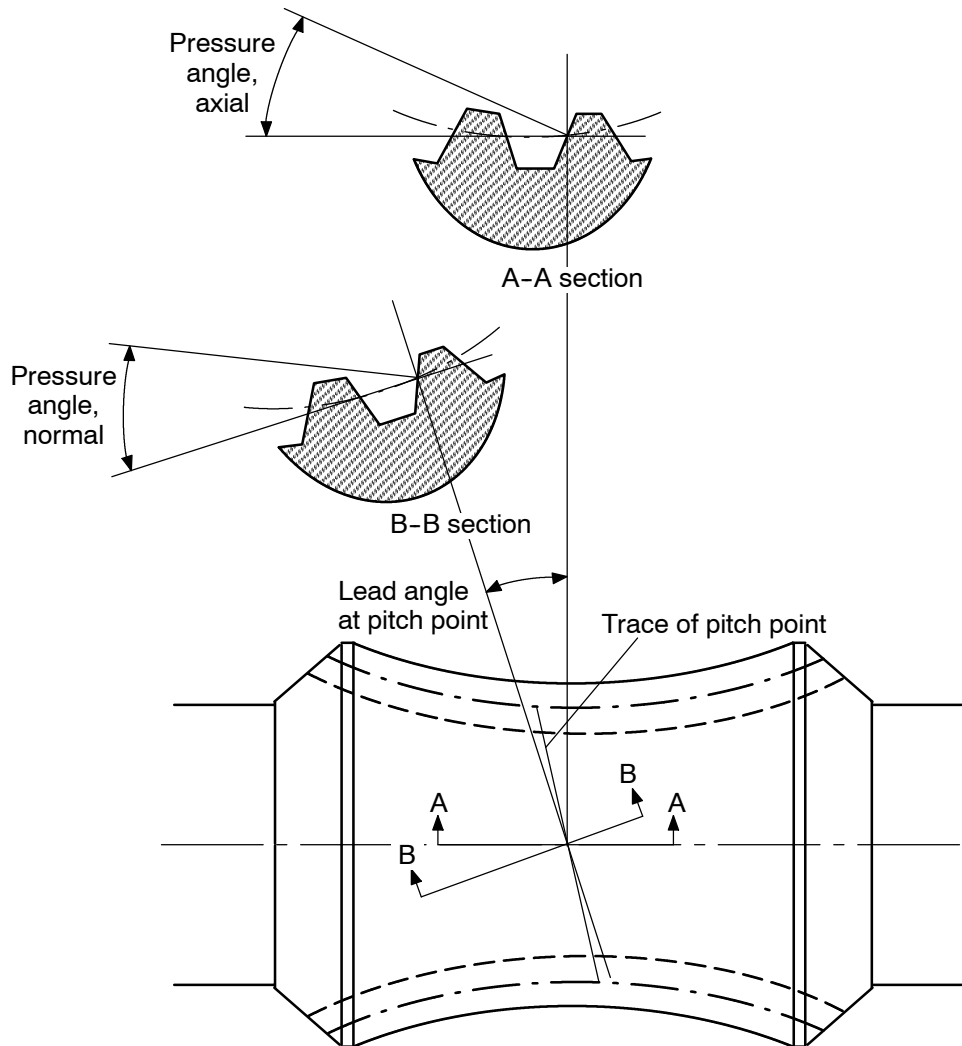


Figure 6 - Lead angle, normal and axial pressure angles

pitch circle, worm: The worm pitch circle is the reference circle used for determination of worm thread proportions. The worm pitch circle lies in the transverse plane. See figures 4 and 5.

pitch diameter, gear, d_{w2} : The gear pitch diameter is the diameter of the gear pitch circle. See figure 4.

pitch diameter, worm, d_{w1} : The worm pitch diameter is the diameter of the worm pitch circle. See figure 2.

pitch point: The pitch point is the common point of the worm pitch circle and the gear pitch circle and is on the line of centers. See figures 4 and 5.

pressure angle, axial, α_x : The axial pressure angle is the pressure angle in the central plane. It is the angle at the pitch point formed between the projection of the normal of the worm thread surface in the central plane and a line parallel to the worm axis in the central plane. See figure 6.

pressure angle, normal, α_n : The normal pressure angle is the pressure angle in the normal plane. It is the angle at the pitch point formed between the normal of the worm thread surface and the tangent plane of the pitch surface. See figure 6.

ratio, μ : The gearset ratio is the number of teeth in the gear divided by the number of threads in the worm.

root circle, gear: The gear root circle coincides with the bottoms of the gear tooth spaces and is defined in the central plane. See figures 1 and 2.

root circle, worm: The worm root circle coincides with the bottom of the worm thread space and is defined in the transverse plane. See figure 4.

root diameter, gear, d_{f2} : The gear root diameter is the diameter of the gear root circle. See figure 4.

root diameter, worm, d_{f1} : The worm root diameter is the diameter of the worm root circle. See figure 2.

thread: The thread is the tooth element of the worm. It forms a helix along the worm face width. The thread advances one or more axial pitches for each worm revolution, with additional threads required for each additional pitch advance.

thread thickness, axial, s_{mx1} : The axial worm thread thickness is the length of arc between two sides of a worm thread measured on the gear pitch circle. See figure 1.

thread thickness, normal, s_{n1} : The normal worm thread thickness is the length of arc between two

sides of a worm thread in a plane normal to the thread surface passing through the pitch point at the center of the thread.

throat diameter, gear, d_{a2} : The gear throat diameter is the diameter of the gear addendum circle. See figure 4.

throat diameter, worm, d_{a1} : The worm throat diameter is the diameter of the worm addendum circle. See figure 2.

throat form radius, gear, r_{g2} : The gear throat form radius is the radius of the throat surface of the gear blank in the transverse plane. See figure 4.

throat form radius, worm, r_{g1} : The worm throat form radius is the radius of the throat surface of the worm blank in the central plane. See figure 2.

tooth thickness, axial, s_{mx2} : The axial gear tooth thickness is the length of arc between the two sides of a gear tooth measured on the gear pitch circle. See figure 1.

tooth thickness, normal, s_{n2} : The normal gear tooth thickness is the length of arc between the two sides of a gear tooth in a plane normal to the tooth surface passing through the pitch point at the center of the gear tooth.

transverse plane: The transverse plane is the plane perpendicular to the worm axis, and contains the gear axis and the line of centers. See figures 3 and 4.

whole depth, h_1, h_2 : The whole depth is the total depth of a tooth space or thread space. It is the radial distance between the addendum circle and the root circle. See figure 1.

working depth, h_w : The working depth is the depth of engagement of the worm thread and gear tooth. See figure 1.

worm: The worm is the threaded input member. The worm has one or more threads, also called leads or starts. The threaded area of the worm has a throat which increases in diameter from the transverse plane toward both ends generally conforming to the curvature of the gear. See figures 2 and 3.

worm axis: The worm axis is the axis of rotation of the worm. The worm axis lies in the central plane. See figures 2 and 3.

wormgearing: Wormgearing refers to gearsets consisting of a worm and a gear.

3 General design considerations

Successful application of globoidal wormgearing begins with an understanding of the unique characteristics of this type of gearing and a consideration of the requirements for the application. This clause discusses both the Hindley and full contact designs.

3.1 Characteristics of wormgearing

Globoidal wormgearing consists of a threaded input member (worm) driving a larger toothed gear at right angles. By restraining the rotating worm axially, the gear is made to turn. As the worm rotates through the gear tooth, lines of contact progress from the outer edges of the face width of the gear tooth toward the center where they exit the mesh. As shown in figures 7 and 8, these contact lines are approximately perpendicular to the direction of rotation of the worm providing a favorable lubrication condition. At any given instant there will be several teeth in contact transmitting the power, where the number of teeth in contact is a function of pitch (tooth size) and worm length. Finer pitches will increase the number of teeth in contact but decrease the working depth. The worm length is limited by the pressure angle where the outermost thread cannot wrap around the gear to a point of interference with the gear teeth.

The Hindley gearset design has approximately 1/8 of the gear teeth in contact with the worm thread at any given instant. The loaded gear teeth are in varying positions along the worm thread as shown in figure 7. At any instant in time, each loaded tooth has one or two contact lines on its face. The contact lines sweep across the entering half of the gear tooth face and converge at the tooth centerline.

The full contact gearset design has approximately 1/8 of the gear teeth in contact with the worm thread at any given instant. The loaded gear teeth are in varying positions along the worm thread as shown in figure 8. At any instant in time, each loaded tooth has two contact lines on its face. The contact lines originate from each side of the gear tooth and progress toward the center where they converge. This design makes full use of the gear tooth by spreading the contact stress over a wider area.

The movement of lines of contact as the tooth progresses through the mesh results in an area of contact marking on each gear tooth. The stresses developed along the lines of contact are surface and subsurface stresses related to the load, the relative

radii of curvature of the worm thread and gear tooth, the distribution of load between the teeth, and the sliding friction forces caused by the rotation of the worm and the materials. When designing gearsets, consideration should be given to maximizing both the total length of contact lines and the relative radii of curvature between the worm and gear.

3.2 Application

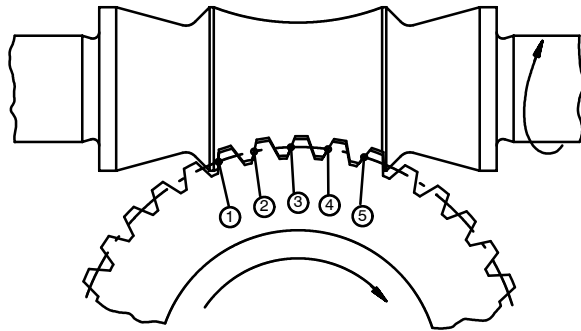
The design of globoidal wormgearing may be tailored to the type of load and speed being transmitted. For transmission of power at higher speeds, a relatively fine pitch gear tooth should be used, while high torques at low speeds require larger worm diameters and coarser pitches. When smoothness of transmission with low noise and vibration is a requirement, finer pitches are preferred. For general design when the application is not predefined, a balanced design approach would be appropriate.

3.3 Ratios

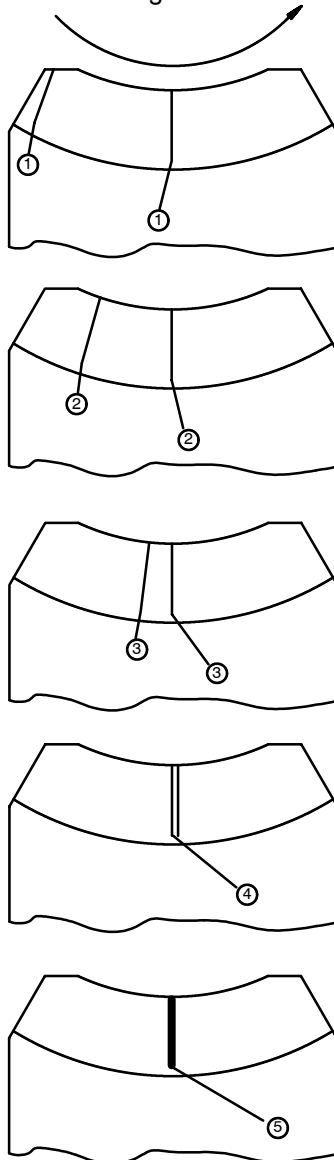
Globoidal wormgearing is generally limited to ratios of 3:1 through 100:1; however, ratios outside this range are possible when due consideration is given to the design and application. The ratio is the number of teeth in the gear divided by the number of threads in the worm. Ratios above 30:1 are usually produced with a single-thread worm and ratios lower than 30:1 favor the use of multiple threads. Generally, the larger the center distance, the greater the number of threads that should be selected. However, there can be exceptions.

In the design or selection of ratios requiring multiple thread worms, there are two tooth combination systems that can be used. One is the even number system where the number of teeth in the gear can be evenly divided by the number of threads in the worm, for example $40/2 = 20:1$. The hunting tooth ratio system uses a combination of gear teeth and worm threads where the number of teeth in the gear is not evenly divisible by the number of threads of the worm, for example $39/2 = 19.5:1$. Over the years, there has been a difference of opinion of the benefit of one system over the other. Use of state-of-the-art manufacturing equipment, tools, and technology has resulted in either of the two systems performing suitably in virtually all types of applications.

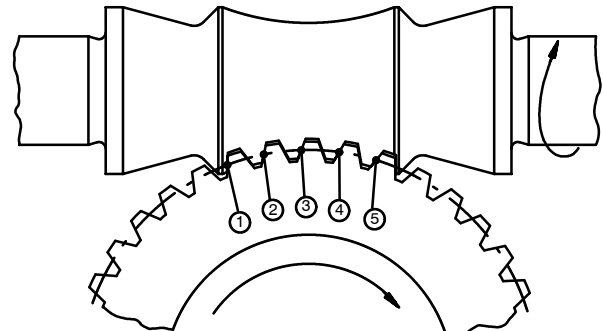
In designs requiring precision motion transmission, or in designs that are extremely cost sensitive, further investigation of the hunting tooth system versus even number system may be required.

**Gear teeth in mesh**

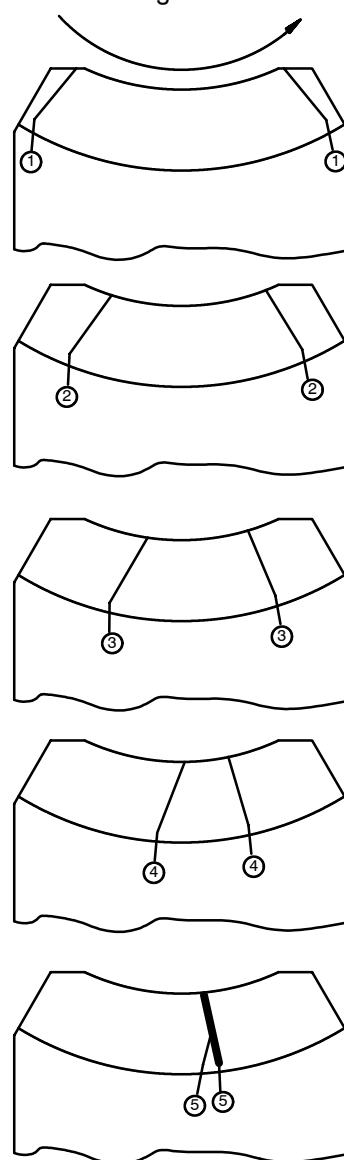
Worm thread
sliding direction



Lines of contact on
successive gear teeth

Figure 7 - Contact lines - Hindley design**Gear teeth in mesh**

Worm thread
sliding direction



Lines of contact on
successive gear teeth

Figure 8 - Contact lines - Full contact design

3.4 Accuracy

Satisfactory performance of wormgearing depends on the accuracy of manufacturing even when an application has no special requirements for indexing, noise level or accuracy.

3.4.1 Worm accuracy

The accuracy of worm dimensions (i.e., runout, lead variations, profile variations, and thread spacing variations for multiple worms) are most important because the worm is the harder member and, as such, will not correct for these variations during run-in. In addition, the worm is subject to many more stress cycles than the gear.

3.4.2 Gear accuracy

The gear is usually made from the softer material and, as such, any initial minor tooth spacing variations as well as minor profile variations will be greatly reduced within a few hours of operation. However, excessive gear tooth runout will tend to accelerate the gear every half revolution causing contact area variations and increased stresses. Excessive face runout or wobble, either in hobbing or mounting of the gear, will also cause the gear contact area to shift across the face, causing the gear to accelerate and decelerate during each revolution.

3.5 Backlash

Gearsets have backlash (clearance) to allow for free running with lubrication between the worm threads and gear teeth. Backlash also allows for expansion of the worm threads and the gear teeth during operation, and for manufacturing tolerances of the worm and gear including tooth thickness and runout.

Backlash should be kept to a minimum in the design of the assembled gearset to minimize dynamic loading of the gear teeth and worm threads in load reversal applications. Gearset backlash will increase slightly after initial operation and run-in (see clause 7) and gradually over the life of the gearset as wear occurs.

3.5.1 Gearset backlash

Gearset backlash is defined as the rotational gear movement at a specified radius with the worm and gear on designed center distance, properly positioned and the worm held stationary. This value is generally converted to arc minutes or degrees. See table C.1 in annex C.

Backlash of worm gears is not defined as holding the gear stationary and rotating the worm because the amount of worm rotation becomes a function of ratio, and is not indicative of the clearance between the worm threads and gear teeth. The worm rotation resulting from gearset backlash is easily calculated by multiplying the gearset backlash measured at the gear (in arc minutes or degrees) by the ratio of the gearset. See annex C for backlash measurement.

Gearset backlash is usually specified by the manufacturer as a range (minimum to maximum) in units of arc minutes or degrees, and varies depending on the size of the gearset and the manufacturing processes used to produce the worm and gear. The backlash is specified and measured on the designed center distance with the worm and gear correctly positioned (see annex A). Each manufacturer specifies the backlash ranges for their various product lines and generally includes this information in their product catalog. Manufacturers may offer a range of options for reduced backlash.

The purchaser often specifies the maximum acceptable gearset backlash for a specific application.

3.5.2 Assembled backlash

When a gearset is assembled into a reducer or as part of a machine, the resulting rotational movement of the gear with the worm held stationary (assembled backlash) will be affected by the gearset backlash, manufacturing tolerances, and design clearances of the housing and mounting components. These include:

- Worm bearing endplay: The bearing endplay or clearance for tapered roller bearings results in axial and radial movement of the worm. Bearing endplay adds directly to the backlash of the gearset to increase assembled backlash. Angular contact (ball) bearings usually do not have endplay.
- Gear bearing endplay: Tapered roller bearings on the gearshaft operate at slow speed and are generally set with zero clearance to slightly preloaded. Even if there is a slight amount of endplay in the gear bearings, it will have very little effect on reducer backlash.
- Housing center distance: The manufacturing tolerance on housing center distance will affect backlash. If the housing center distance is greater than the design center distance (open condition), reducer assembled backlash will increase. If it is

less than the design center distance (closed condition), assembled backlash will decrease.

- Fits between bearing supports and bearings: If the worm or gear bearing outer race is mounted in the housing bore, there is a slight clearance between the housing bore and the outside diameter of the bearing race for assembly purposes. If the worm or gear bearing is mounted in a bearing carrier, the outer race is generally a press fit in the bore of the carrier, but there is a slight clearance between the outside diameter of the bearing carrier and the housing bore for assembly purposes. These clearances allow the worm and gear to separate slightly (increase center distance) under load increasing the backlash of the reducer.
- Worm and gear bearing runout: Runout in the bearings results in slight opening and closing of the gearset center distance and a corresponding increase and decrease in assembled backlash for each revolution of the worm and gear.
- Worm, gear and gearshaft runout: Runout between the worm pitch circle and the worm bearing diameters, runout between the gear pitch circle and the gear bore, and runout between the gear diameter and bearing diameters on the gearshaft will all slightly add to or subtract from assembled backlash during rotation of the components as the center distance opens or closes.
- Temperature: When the reducer is operating under load, the worm, gear, housing and related components all expand as the reducer comes up to operating temperature. Although each component reacts differently because of nonuniform temperature distribution throughout the reducer, if the reducer operates continuously at a stabilized temperature, most of the steel and cast iron components will expand about the same amount and operating clearances will be similar to what they were when the unit was cold. However, the bronze gear will expand approximately 50% more than the cast iron and steel components, which may result in a slight decrease in assembled backlash with the unit at operating temperature.

Each of the above items will affect the assembled backlash. However, it is highly unlikely that all of the tolerances and fits will combine to increase backlash to the maximum possible or decrease backlash to the minimum possible amount. Speed reducer manufacturers generally publish nominal as-

sembled reducer backlash values based on mid-range tolerances.

Assembled backlash is measured by: locking the worm so that it cannot rotate, yet leaving it free to move axially; rotating the gearshaft in both directions through the backlash; and, using a dial indicator to determine the amount of gearshaft movement in millimeters at a measured radius. When rotating the gearshaft, it is important to apply sufficient torque to the gearshaft in both directions to “seat” the worm bearings and obtain an accurate measurement of backlash. See annex C for additional information on measuring backlash.

3.5.3 Low backlash

Many manufacturers offer gearsets or reducers with reduced backlash, near zero and zero backlash for applications requiring accurate positioning, or a minimum of “lost motion” when reversing.

Low backlash gearsets are manufactured with less backlash than standard products, by increasing tooth or thread thickness. Backlash may be specified by the manufacturer or the purchaser.

Backlash increases with gear wear. Low backlash gearsets or reducers should be selected using a service factor at least 20% higher than normal to minimize wear.

3.5.4 Low backlash reducers

Low backlash reducers are equipped with a low backlash gearset as described above and may include the following features:

- Reduced worm bearing endplay: A reduction in worm bearing endplay provides an equivalent reduction in reducer backlash measured at the gear pitch circle. When using tapered roller bearings, the standard endplay may be reduced depending on reducer duty cycle, speed, loading and operating temperature. When using angular contact (ball) bearings, endplay can be reduced to zero.
- Precision bearings: Bearings with reduced runout will reduce the backlash variation of the speed reducer.
- Closing center distance: Closing the center distance of a low backlash gearset slightly will decrease reducer backlash. Attempts to reduce the center distance by more than a small amount will distort the gearset contact and cause increased wear. Reducing the center distance can also result in “bottoming” the throat of the worm or gear in the root of the mating part.

3.5.5 Zero backlash gearsets and reducers

These products are highly engineered and each application must be reviewed with the manufacturer:

- Split worms: The worms for these gearsets are split through the transverse plane with the floating half of the worm threads preloaded through the worm bearings against the gear teeth. This arrangement provides zero backlash and accommodates wear as it occurs in the gearset and worm bearings. Additional input torque is required to overcome the preload between the worm and gear.
- Split gears: The gears are split through the central plane with the floating half of the gear preloaded against the worm threads. This arrangement provides zero backlash and accommodates wear as it occurs in the gearset. Additional input torque is required to overcome the preload between the worm and gear.

4 Gearset design

Design data is presented as steps arranged in the same order that a designer would normally use to calculate dimensions of the gearing. It is recognized that these steps will not always be followed in the same sequence, being dependent on the known factors and requirements for the gearing. The ratio and center distance should be established prior to the start of the design.

If a new gearset is being designed as part of a standard product line, the dimensions of the standard worm and gear blank should be considered when determining gearset proportions.

The following design information is presented to establish the basic worm and gear proportions for a globoidal gearset. All of the design data applies to the Hindley design. Some of this information is also used to calculate worm and gear proportions for the full contact design.

4.1 Number of teeth in gear, z_2

The number of teeth in the gear is influenced to a large extent by center distance. In order to achieve proper tooth proportions, the number of teeth in the gear should be selected from table 2. Special applications may require selecting a different number of gear teeth. The maximum number of gear teeth for single-threaded worms is limited only by manufacturing considerations.

Table 2 - Recommended number of gear teeth

Center distance, mm		Minimum number of gear teeth	Maximum number of gear teeth with multiple threaded worms
From	Less than		
38	- 65	24	40
65	- 90	24	50
90	- 150	30	50
150	- 330	40	60
330	- 480	50	60
480	- 585	50	70
585	- Larger	60	80

4.2 Number of threads in worm, z_1

The number of teeth in the gear and the ratio will determine the number of threads for the worm. Generally, one to ten threads are used. In special cases a larger number of threads may be required.

$$z_1 = \frac{z_2}{u} \quad (1)$$

where

- z_1 is the number of threads in worm;
- z_2 is the number of teeth in gear;
- u is the gear ratio.

4.3 Worm pitch diameter, d_{w1}

The first approximation of worm pitch diameter is:

$$d_{w1} = \frac{a^{0.875}}{1.47} \quad (2)$$

where

- d_{w1} is the worm pitch diameter in transverse plane, mm;
- a is the center distance, mm.

This value may be adjusted after calculating the worm root stress.

4.4 Minimum worm root diameter, d_{f1}

$$d_{f1} = \frac{a^{0.875}}{2} \quad (3)$$

where

- d_{f1} is the worm root diameter in transverse plane, mm.

4.5 Gear pitch diameter, d_{w2}

$$d_{w2} = 2a - d_{w1} \quad (4)$$

where

- d_{w2} is the gear pitch diameter in central plane, mm.

4.6 Circular pitch, p_{t2}

$$p_{t2} = \frac{\pi d_{w2}}{z_2} \quad (5)$$

where

p_{t2} is the circular pitch, mm.

4.7 Axial module, m_a

$$m_a = \frac{d_{w2}}{z_2} \quad (6)$$

where

m_a is the axial module, mm.

4.8 Lead angle at the pitch point of the worm, δ_{m1}

$$\delta_{m1} = \tan^{-1} \frac{d_{w2}}{d_{w1} u} \quad (7)$$

where

δ_{m1} is the lead angle at pitch point, degrees.

4.9 Normal circular pitch, p_{n1}

$$p_{n1} = p_{t2} \cos \delta_{m1} \quad (8)$$

where

p_{n1} is the normal circular pitch, mm.

4.10 Normal module, m_n

$$m_n = m_a \cos \delta_{m1} \quad (9)$$

where

m_n is the normal module, mm.

4.11 Whole depth, $h_{1,2}$

$$h_1 = h_2 = \frac{p_{n1}}{2} \quad \text{or} \quad (10)$$

$$h_1 = h_2 = \frac{\pi m_n}{2}$$

where

h_1 is the whole depth of the worm thread, mm;

h_2 is the whole depth of gear tooth, mm.

The value obtained in this equation is a starting point and may be varied to suit the design. The whole depth of the worm is generally equal to the whole depth of the gear.

4.12 Working depth worm and gear, h_w

$$h_w = 0.9 h_1 = 0.9 h_2 \quad (11)$$

where

h_w is the working depth of tooth, mm.

4.13 Addendum worm and gear, $h_{am1,2}$

$$h_{am1} = h_{am2} = \frac{h_w}{2} \quad (12)$$

where

h_{am1} is the addendum of the worm thread, mm;

h_{am2} is the addendum of the gear tooth, mm.

4.14 Clearance worm and gear, $c_{1,2}$

$$c_1 = h_1 - h_w \quad (13)$$

$$c_2 = h_2 - h_w \quad (14)$$

where

c_1 is the clearance in the worm, mm;

c_2 is the clearance in the gear, mm.

4.15 Dedendum worm and gear, $h_{fm1,2}$

$$h_{fm1} = h_1 - h_{am1} \quad (15)$$

$$h_{fm2} = h_2 - h_{am2} \quad (16)$$

where

h_{fm1} is the dedendum of worm thread, mm;

h_{fm2} is the dedendum of gear tooth, mm.

4.16 Worm throat diameter, d_{a1}

$$d_{a1} = d_{w1} + 2 h_{am1} \quad (17)$$

where

d_{a1} is the worm throat diameter in transverse plane, mm.

4.17 Worm root diameter, d_{f1}

$$d_{f1} = d_{w1} - 2h_{fm1} \quad (18)$$

The root diameter obtained in this equation should be compared to the value obtained from equation 3. If the value obtained from equation 18 is less than the minimum, reducing the gear pitch diameter until the worm root diameter meets the minimum requirement, or increasing the number of gear teeth, should be considered.

In order to obtain optimum tooth proportions, it is desirable that the worm root diameter should not be less than that indicated by the above equation. For ratios less than 8:1, the worm root diameter may be increased. This increase may vary up to 15 percent for a 3:1 ratio, and is done for manufacturing purposes.

NOTE: When the final worm root diameter is determined, the combined bending and torsional stress must be calculated to insure adequate strength.

4.18 Normal pressure angle, α_n

The normal pressure angle is generally 20 to 25 degrees.

4.19 Axial pressure angle, α_x

$$\alpha_x = \tan^{-1} \frac{\tan \alpha_n}{\cos \delta_{m1}} \quad (19)$$

where

α_x is the axial pressure angle, degrees;

α_n is the normal pressure angle, degrees.

4.20 Worm thread and gear tooth thickness

It is common practice in globoidal wormgearing to proportion the gear tooth and worm thread thickness as follows: the axial gear tooth thickness, s_{mx2} , is generally 55 percent of the circular pitch, and the axial worm thread thickness, s_{mx1} , is 45 percent of the circular pitch. The backlash of the gearset is subtracted from the axial worm thread thickness. This practice will produce greater tooth strength in the gear, which is the limiting member. See annex C for backlash.

4.21 Base diameter for Hindley design only, d_b

$$d_b = d_{w2} \sin \left(\alpha_x + \frac{180 s_{mx2}}{p_{t2} z_2} \right) \quad (20)$$

where

d_b is the base diameter (Hindley design), mm;

s_{mx2} is the axial gear tooth thickness, mm.

The base circle can be determined using equation 20 or constructed from a layout as shown in figure 2. The base circle lies in the central plane. After laying out the centerline of the worm and gear, the line of centers and the gear pitch circle, measure along the common worm and gear pitch circle to the right or left of the line of centers an amount equal to 27.5 percent of the circular pitch, p_{t2} . Through the point thus established, and at an angle to the line of centers equal to the axial pressure angle, α_x , extend a line upward. A circle tangent to this line and concentric to the gear axis is the base circle.

4.22 Tooth or thread profile for Hindley design only

The thread profile is usually straight in the central plane, but any other form may be used. Because there is no rolling action up or down the flanks, the profile is unimportant except that it must be the same on the worm and hob. The straight-sided tooth provides the greatest ease of manufacture and checking, for both gearsets and cutting tools.

4.23 Effective worm thread length for Hindley design only, b_{1eff}

The effective length of the worm thread should be the base diameter, minus approximately 0.02 times center distance. The worm thread extending beyond the effective length must be relieved to prevent interference. The effective length cannot extend outside of the base circle. Significantly reducing the

effective length of the worm thread will reduce the rating of the gearset.

Effective worm thread length

$$b_{1eff} = d_b - 0.02 a \quad (21)$$

where

b_{1eff} is the effective worm thread length, mm.

4.24 Worm outside diameter, d_{e1}

The worm outside diameter is generally scaled from the gearset layout. If a worm will be manufactured from a standard blank, this also must be considered in the gearset design.

4.25 Worm flat, b_{f1}

$$b_{f1} = \frac{p_{t2}}{5.5} \quad (22)$$

or,

$$b_{f1} = \frac{\pi m_a}{5.5} \quad (23)$$

where

b_{f1} is the length of flat on outside diameter of worm, mm.

4.26 Gear face width, b_2

The face width of the gear is generally equal to or slightly less than the root diameter of the worm. Additional face width will not increase the rating, and is wasteful of material. Conversely, significant reduction of face width will reduce gearset rating.

4.27 Gear root form radius, r_{f2}

To construct the gear root form radius, add twice the clearance to 0.5 times the worm throat diameter and lay this distance out along the centerline of the gear from the root of the gear tooth. Draw a radius from this point to the gear faces.

Gear root form radius

$$r_{f2} = 2c_2 + 0.5 d_{a1} \quad (24)$$

where

r_{f2} is the gear root form radius, mm.

4.28 Gear face angle

The gear face angle is typically 30° – 35°. If the gear will be manufactured from a standard blank, this must also be considered in the design.

4.29 Worm face angle

The worm face angle is typically 30° – 45°. If the worm will be manufactured from a standard blank, this must also be considered in the design.

4.30 Worm throat form radius, r_{g1}

The worm throat form radius is generally selected by judgement when the worm outside diameter is scaled from the gearset layout. If a worm will be manufactured from a standard blank, this also must be considered. An approximate value for the worm throat form radius is:

$$r_{g1} = 1.06 \left(a - \frac{d_{a1}}{2} \right) \quad (25)$$

where

r_{g1} is the worm throat form radius, mm.

4.29 Gear throat form radius, r_{g2}

The gear throat form radius is struck from the same point as the gear root form radius, but is shorter in length by one whole depth.

$$r_{g2} = r_{f2} - h_2 \quad (26)$$

where

r_{g2} is the gear throat form radius, mm.

5 Self-locking

A gearset is said to be self-locking or non-backdriving when torque applied at the low speed shaft will not result in motion from a static condition. All gearsets will backdrive to some degree if they are in motion. It is not accepted practice to consider any worm gearset to be statically self-locking.

A test of a particular gearset in its actual operating environment will not provide complete assurance of self-locking since, after running, the gear teeth will become polished, increasing the probability for backdriving. *To ensure irreversibility, a suitable brake must be used.*

6 Adjustments and contact patterns of globoidal wormgearing

Globoidal gearsets must be positioned properly at assembly to obtain the desired contact between the worm threads and gear teeth. Gearset mountings should allow for adjustment of the worm in the worm axial direction (worm end position), and the gear in the gear axial direction (gear side position).

Figures 9 through 11 illustrate examples of correct contact patterns before and after run-in, and incor-

rect contact patterns with the worm off end position and the gear off side position. The contact patterns for Hindley design gearsets vary with ratio. The 10:1 ratio and 30:1 ratio Hindley designs are shown in figures 9 and 10. Contact patterns for full contact design gearsets do not vary significantly with ratio. A 30:1 ratio full contact design is shown in figure 11.

7 Run-in procedures

The running-in of globoidal wormgearing immediately after installation of new gearing may not always be necessary, but doing so is certainly advantageous.

When the gearing is correctly designed for its load conditions and properly manufactured, there is generally no difficulty if its full load is applied from the start. However, there are advantages to be gained from an initial running under part load, followed by a gradual build-up to full load.

7.1 Results of run-in

No matter how well finished wormgearing surfaces may be, they do require some time to run in to conform and achieve a work hardened surface on the bronze. The gearing has a better chance of providing maximum performance if the initial working can be done gradually.

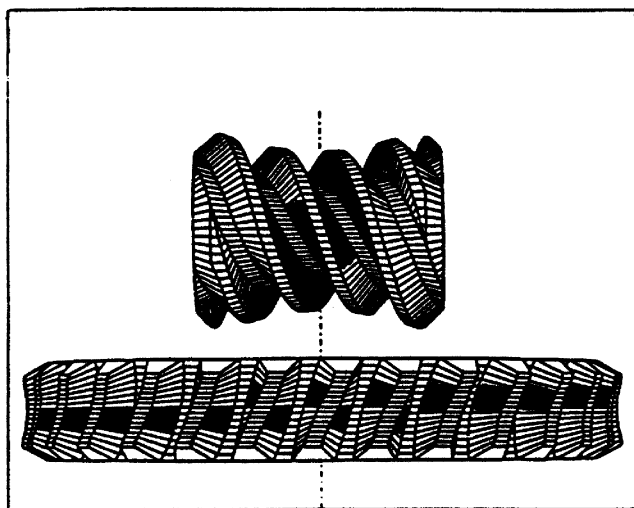
7.2 Initial running at full load

An immediate application of full load or overloads, due to new driven equipment, concentrates high contact pressures on small areas which may cause some temporary damage to the surfaces and may cause high local surface temperatures. The temporary damage to the gear bronze surfaces will often "heal" after continued running at full or less than full load. The run-in of wormgearing at gradually increasing loads can prevent the occurrence of such surface damage.

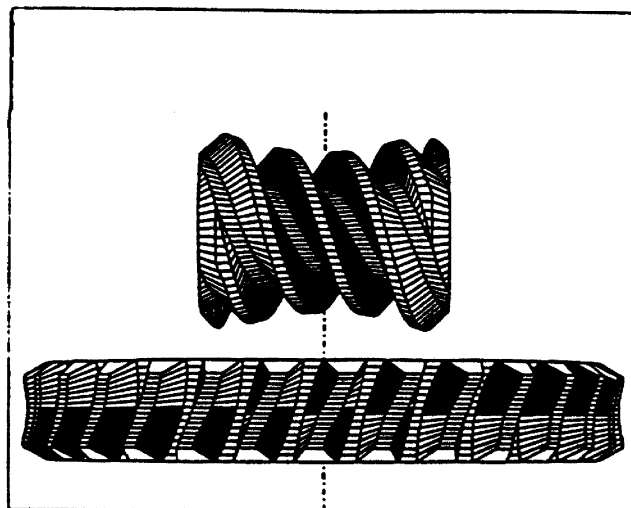
7.3 Recommended run-in period

Experience of testing and running of wormgearing indicates that the initial friction is 10 to 15 percent higher than that which will be obtained after run-in is completed. The first few hours of operation at gradually increasing loads reduces the friction and the efficiency settles down to a steady figure after about 10 - 100 hours of operation, depending on the size and speed of the gearing.

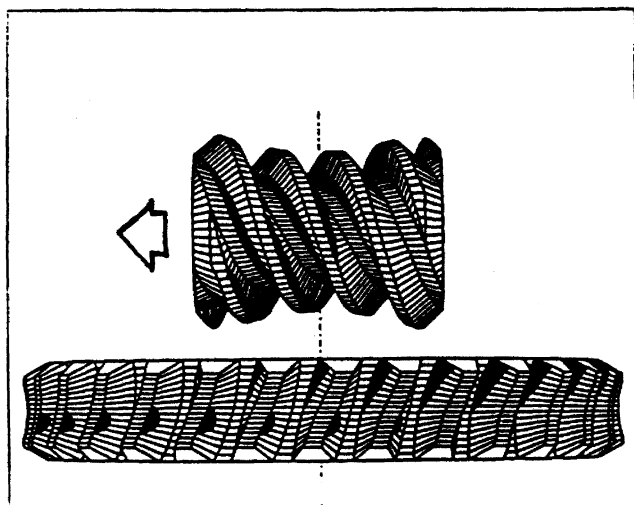
A reasonable run-in procedure is to apply approximately one-half load for a few hours and then to increase the load in at least two stages.



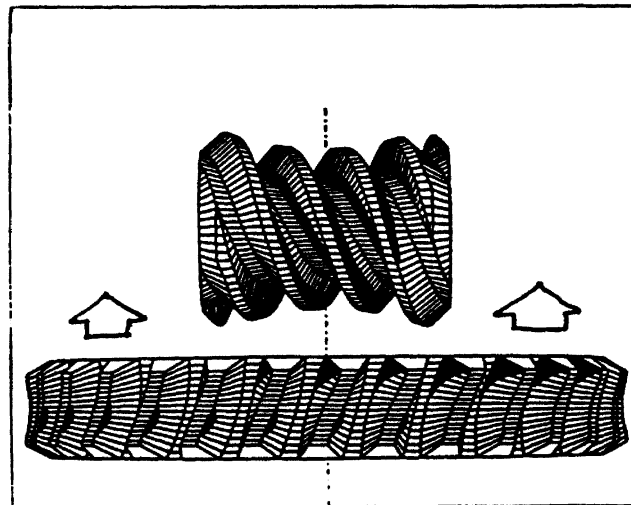
Correct contact pattern before running-in.



Correct contact pattern after running-in.



How to correct for "worm off end position".



How to to correct for "gear off side position".

Figure 9 - Hindley design, 101.60 mm C.D., 10:1 ratio, 3 threads, 30 teeth

7.4 Gear pitting

The surface damage occurring during initial running, whether at full load or reduced load, and described in 7.2 may result in scattered corrective pitting on the gear tooth surfaces after 500 - 1000 hours of additional operation. Laboratory load testing of even extensively pitted bronze gears has shown that accuracy of motion transmission, gearing efficiency, noise level, load capacity, and the heat generated may not be significantly affected by the presence of such pitting.

8 Rating and efficiency

The rating procedure in clause 8 is applicable to globoidal wormgearing designed in accordance with

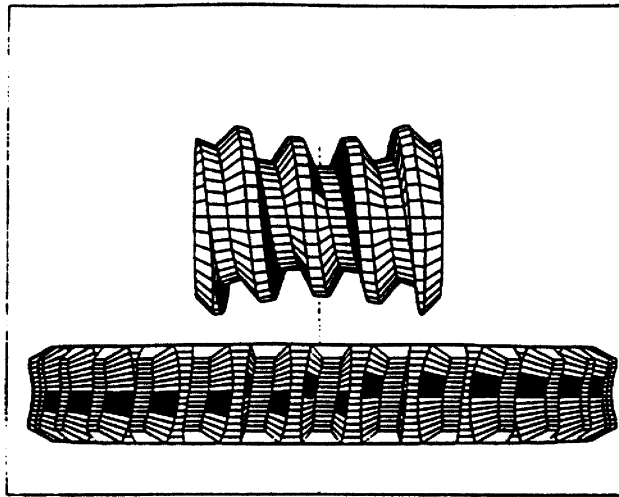
clause 4 of this standard. This procedure is taken from ANSI/AGMA 6017-E86 except in metric format.

8.1 Input power rating of gearing

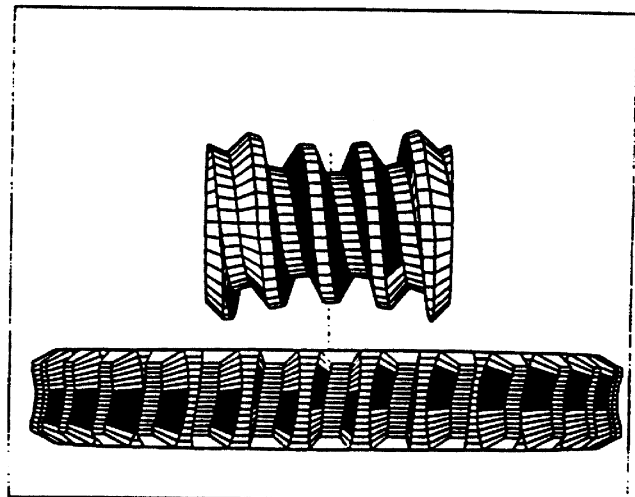
8.1.1 Type of service

The input power rating formula contained in this section develops a power value applicable to a unity (1.0) service factor. A service factor of 1.0 applies to the following operating conditions:

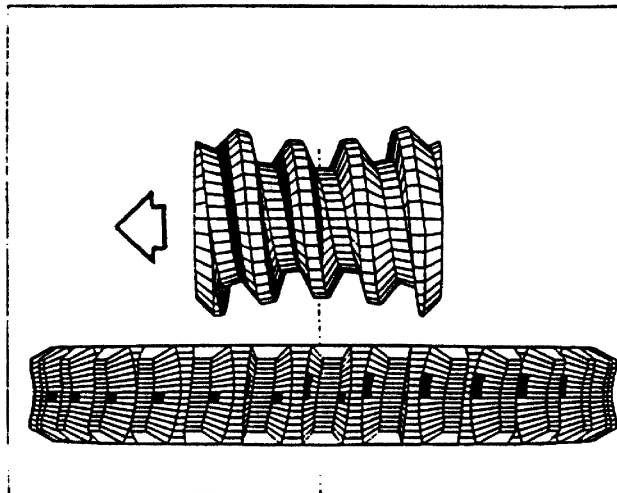
- uniform load, free of shock pulses;
- continuous service up to 10 hours per day.



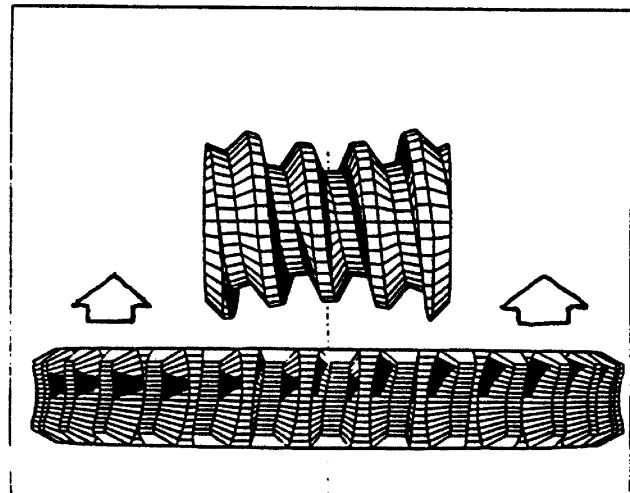
Correct contact pattern before running-in.



Correct contact pattern after running-in.



How to correct for "worm off end position".



How to correct for "gear off side position".

Figure 10 - Hindley design, 101.60 mm C.D., 30:1 ratio, 1 thread, 30 teeth

8.1.2 Starting loads and momentary overloads

Starting loads and momentary overloads up to 300 percent of the input power rating are permissible. The reducer and its components must be designed for this 300% overload condition. Loads in excess of 300% can result in reducer failure.

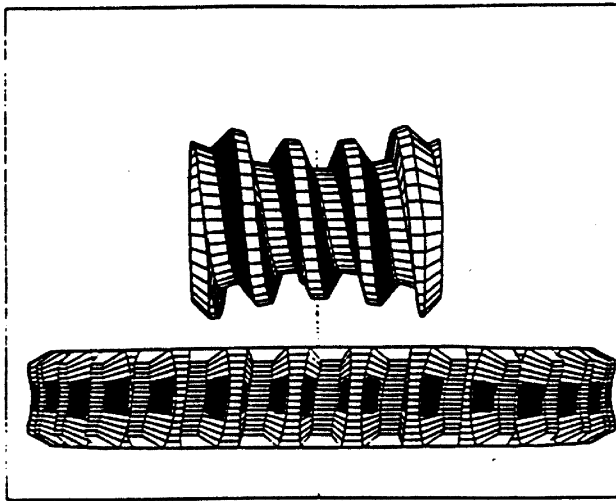
8.1.2.1 Duration of starting loads and momentary overloads

These loads are applied to a reducer for one to two

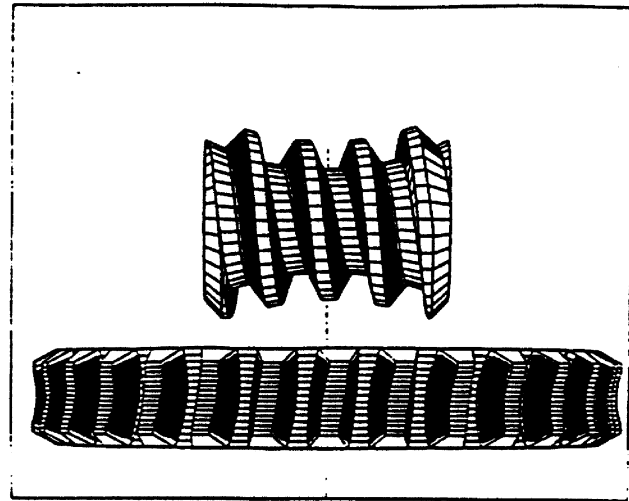
seconds duration and may occur when the reducer is starting, running, stopping, or in a static condition.

8.1.2.2 Frequency of application

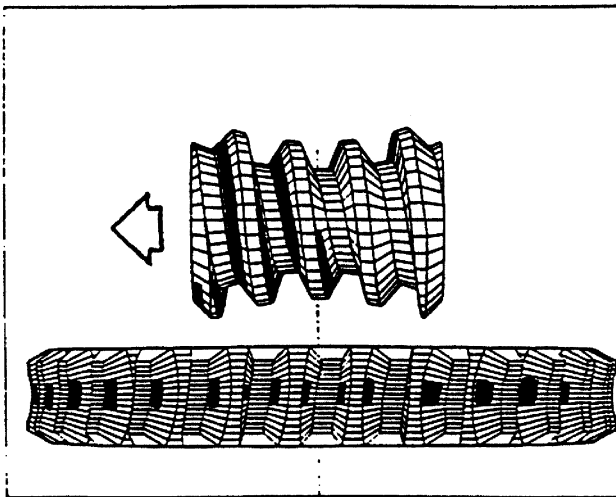
Since momentary overloads can result in stresses at the mesh and in other reducer components that exceed the endurance limit of the materials used, the total number of applications of momentary overloads should not exceed 25 000 over the life of the reducer.



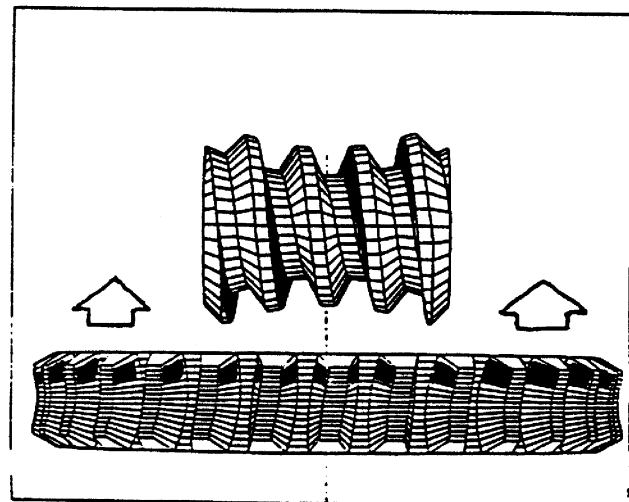
Correct contact pattern before running-in.



Correct contact pattern after running-in.



How to correct for "worm off end position".



How to to correct for "gear off side position".

Figure 11 - Full contact design, 101.60 mm C.D., 30:1 ratio, 1 thread, 30 teeth

8.1.2.3 Life

When the appropriate service factor is selected, and proper lubrication and maintenance practices are followed, experience shows that nominal wormgear service life of 25 000 hours will be attained. If a service life of greater than 25 000 hours is required, a higher service factor than specified in annex D, shall be used. If a lower service life than 25 000 hours is acceptable, a lower service factor than specified in annex D can be used if the manufacturer is consulted about the particular gearing being used.

8.1.3 Fundamental rating formula for surface durability

The equation for input power rating includes factors for ratio, pressure, size, face width, material, and sliding velocity of the worm. The equation is:

$$P_1 = \frac{0.746 n_1}{u} Z_w Z_u Z_m Z_v \quad (27)$$

where

P_1 is the input power rating, kW;

n_1 is the rotational speed of worm, rpm;

Z_w is the basic pressure factor;

Z_u is the ratio correction factor;

Z_m is the face width and material factor;

Z_v is the velocity factor.

8.1.4 Empirical factors

Values of empirical factors used in the above equation are obtained as follows:

- basic pressure factor, Z_w , is to be selected from table 3;
- ratio correction factor, Z_u , is to be selected from table 4;
- face width and material factor, Z_m , is to be selected from table 5;
- velocity factor, Z_v , is to be selected from table 6, using sliding velocity, V_g .

$$V_g = \frac{n_1 d_{a1}}{19\,098 \cos \delta_{m1}} \quad (28)$$

where

V_g is the sliding velocity, m/s.

8.2 Input power rating - single and multiple reductions

8.2.1 Rating determination

The input power rating of each stage of a multiple reduction unit must be determined by considering each stage to be separate and connected in series. The input power rating of any stage of a globoidal speed reducer is determined from equation 27. This is important because one stage will be limiting and will establish the allowable input power rating of the reducer.

The input power rating of each reduction must be determined at the input speed of that stage, using the applicable AGMA document for rating that type of gearing. Due consideration must be given to the overload capacity of each reduction based on that type of gearing.

8.2.2 Unit rating

The input rating of the unit is determined by referring each stage of reduction back to the input shaft of the first stage. The input power rating of each stage is

divided by the efficiencies of all preceding stages. Typical efficiencies for globoidal gearing are shown in annex F. The efficiency can vary substantially depending on center distance, reduction ratio and speed. The efficiencies for stages not covered by this standard should be established using the applicable standard as stated in 8.2.1.

The unit rating is the lesser of the following:

- Input power rating (1st stage)
- $\frac{\text{Input power rating (2nd stage)}}{\text{Efficiency (1st stage)}}$
- $\frac{\text{Input power rating (3rd stage)}}{\text{Efficiency (1st stage)} \times \text{Efficiency (2nd stage)}}$

Table 3 - Basic pressure factor, Z_w

Center distance, a mm	Basic pressure factor, Z_w
50	0.0816
65	0.1661
75	0.2446
90	0.4794
100	0.6030
125	1.1030
150	1.6170
175	2.4220
200	3.4780
250	6.3680
300	10.4390
375	19.3930
450	31.8940
550	55.1440
600	68.9900
650	85.7760
700	104.9380
750	126.6060
800	150.9090
850	177.9710
900	205.3670
950	238.5950
1000	275.0750

Table 4 - Ratio correction factor, Z_u

Ratios 2 to 20										
Ratio, u	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
2	0.280	0.290	0.300	0.310	0.320	0.330	0.340	0.350	0.360	0.370
3	0.380	0.391	0.402	0.413	0.424	0.435	0.446	0.457	0.468	0.479
4	0.490	0.496	0.502	0.508	0.514	0.520	0.526	0.532	0.538	0.544
5	0.550	0.555	0.561	0.566	0.572	0.577	0.582	0.588	0.593	0.599
6	0.604	0.607	0.610	0.612	0.615	0.618	0.621	0.624	0.626	0.629
7	0.632	0.635	0.639	0.642	0.645	0.648	0.652	0.655	0.658	0.662
8	0.665	0.666	0.667	0.668	0.669	0.670	0.671	0.672	0.673	0.674
9	0.675	0.676	0.678	0.679	0.681	0.682	0.684	0.685	0.687	0.688
10	0.690	0.691	0.692	0.693	0.694	0.695	0.696	0.697	0.698	0.699
11	0.700	--	0.701	--	0.702	--	0.703	--	0.704	--
12	0.706	--	0.707	--	0.708	--	0.710	--	0.711	--
13	0.712	--	0.713	--	0.715	--	0.717	--	0.718	--
14	0.720	--	0.720	--	0.721	--	0.722	--	0.722	--
15	0.723	--	0.724	--	0.725	--	0.725	--	0.726	--
16	0.727	--	0.727	--	0.728	--	0.728	--	0.729	--
17	0.730	--	0.730	--	0.731	--	0.731	--	0.732	--
18	0.732	--	0.732	--	0.733	--	0.733	--	0.734	--
19	0.735	--	0.735	--	0.736	--	0.736	--	0.737	--
20	0.737	--	--	--	--	--	--	--	--	--
Ratios 20 to infinity										
Ratio, u	0	1	2	3	4	5	6	7	8	9
2	0.737	0.739	0.739	0.740	0.741	0.742	0.743	0.744	0.745	0.745
3	0.746	--	0.746	--	0.747	--	0.747	--	0.748	--
4	0.748	--	0.748	--	0.749	--	0.749	--	0.750	--
5	0.750	--	0.750	--	0.751	--	0.751	--	0.751	--
6	0.751	--	0.751	--	0.752	--	0.752	--	0.752	--
7	0.752	--	0.752	--	0.753 ¹⁾	--	--	--	--	--
NOTE:										
1) For ratios 74:1 and higher, $Z_u = 0.753$.										

Table 5 - Face width and materials factor, Z_m
 (These factors apply only for materials listed in 8.8.4 and 8.8.5.)

Center distance, a mm	Face width and materials factor, Z_m	Center distance, a mm	Face width and materials factor, Z_m
50	0.6157	450	1.3220
65	0.6925	550	1.3770
75	0.7383	600	1.3980
90	0.8012	650	1.4100
100	0.8399	700	1.4210
125	0.9282	750	1.4310
150	1.0070	800	1.4410
175	1.0610	850	1.4500
200	1.1030	900	1.4590
250	1.1780	950	1.4670
300	1.2420	1000	1.4750
375	1.2820		

Table 6 - Velocity factor, Z_v

Values for V_g from 0 thru 5.99 m/s										
V_g	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.750	0.745	0.740	0.735	0.730	0.725	0.721	0.716	0.711	0.706
0.1	0.701	0.696	0.691	0.686	0.683	0.680	0.677	0.673	0.669	0.665
0.2	0.661	0.658	0.655	0.654	0.649	0.646	0.643	0.640	0.637	0.634
0.3	0.632	0.629	0.625	0.622	0.620	0.618	0.616	0.614	0.612	0.610
0.4	0.608	0.606	0.604	0.602	0.600	0.598	0.596	0.594	0.592	0.590
0.5	0.589	0.588	0.587	0.586	0.584	0.583	0.582	0.581	0.580	0.578
0.6	0.577	0.576	0.575	0.574	0.572	0.571	0.570	0.569	0.568	0.566
0.7	0.565	0.564	0.563	0.562	0.561	0.559	0.558	0.556	0.555	0.554
0.8	0.553	0.552	0.551	0.549	0.548	0.547	0.546	0.544	0.543	0.542
0.9	0.541	0.540	0.539	0.537	0.536	0.535	0.534	0.533	0.532	0.531
1.0	0.530	0.529	0.527	0.526	0.525	0.524	0.523	0.522	0.521	0.520
1.1	0.519	0.518	0.517	0.516	0.515	0.513	0.512	0.511	0.510	0.509
1.2	0.508	0.507	0.506	0.504	0.503	0.502	0.501	0.500	0.499	0.498
1.3	0.497	0.496	0.494	0.493	0.492	0.491	0.490	0.489	0.488	0.487
1.4	0.486	0.485	0.484	0.483	0.482	0.480	0.479	0.478	0.477	0.476
1.5	0.475	0.474	0.473	0.471	0.470	0.469	0.468	0.467	0.466	0.465
1.6	0.464	0.463	0.462	0.461	0.460	0.459	0.458	0.457	0.456	0.455
1.7	0.454	0.453	0.452	0.451	0.450	0.449	0.448	0.447	0.446	0.445
1.8	0.444	0.443	0.442	0.441	0.440	0.439	0.438	0.437	0.436	0.435
1.9	0.434	0.433	0.432	0.431	0.430	0.429	0.428	0.427	0.426	0.425
2.0	0.424	0.423	0.422	0.421	0.420	0.419	0.418	0.417	0.416	0.415
2.1	0.415	0.414	0.413	0.412	0.411	0.410	0.409	0.408	0.407	0.406
2.2	0.405	0.404	0.404	0.403	0.403	0.402	0.401	0.400	0.399	0.399
2.3	0.398	0.397	0.396	0.395	0.395	0.394	0.393	0.392	0.391	0.391
2.4	0.390	0.389	0.389	0.388	0.387	0.386	0.385	0.385	0.384	0.383
2.5	0.382	0.381	0.381	0.380	0.380	0.379	0.378	0.377	0.376	0.376
2.6	0.375	0.374	0.373	0.372	0.372	0.371	0.370	0.369	0.369	0.368
2.7	0.368	0.367	0.367	0.366	0.366	0.365	0.364	0.364	0.363	0.363
2.8	0.363	0.362	0.362	0.361	0.360	0.360	0.359	0.359	0.358	0.357
2.9	0.357	0.356	0.356	0.355	0.355	0.354	0.354	0.354	0.353	0.353
3.0	0.352	0.352	0.351	0.351	0.350	0.350	0.349	0.349	0.348	0.348
3.1	0.347	0.347	0.346	0.346	0.345	0.344	0.344	0.343	0.342	0.342
3.2	0.341	0.341	0.340	0.340	0.339	0.338	0.338	0.337	0.337	0.336
3.3	0.336	0.336	0.335	0.335	0.334	0.334	0.333	0.333	0.332	0.332
3.4	0.331	0.331	0.330	0.330	0.329	0.329	0.328	0.328	0.327	0.327
3.5	0.326	0.326	0.326	0.326	0.325	0.325	0.325	0.325	0.324	0.324
3.6	0.323	0.323	0.323	0.322	0.322	0.321	0.321	0.321	0.320	0.320
3.7	0.319	0.319	0.319	0.318	0.318	0.317	0.317	0.317	0.316	0.316

(continued)

Table 6 (continued)

Values for V_g from 0 thru 5.99 m/s										
V_g	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
3.8	0.315	0.315	0.315	0.314	0.314	0.313	0.313	0.313	0.312	0.312
3.9	0.311	0.311	0.311	0.310	0.310	0.309	0.309	0.309	0.308	0.308
4.0	0.307	0.307	0.307	0.306	0.306	0.306	0.305	0.305	0.304	0.303
4.1	0.303	0.302	0.302	0.302	0.301	0.301	0.300	0.300	0.300	0.299
4.2	0.299	0.298	0.298	0.298	0.297	0.297	0.296	0.296	0.296	0.296
4.3	0.295	0.295	0.295	0.295	0.295	0.294	0.294	0.294	0.294	0.293
4.4	0.293	0.292	0.292	0.292	0.292	0.291	0.291	0.291	0.291	0.290
4.5	0.290	0.289	0.289	0.289	0.288	0.288	0.287	0.287	0.287	0.286
4.6	0.286	0.286	0.285	0.285	0.285	0.284	0.284	0.284	0.284	0.283
4.7	0.283	0.283	0.282	0.282	0.282	0.281	0.281	0.281	0.281	0.281
4.8	0.280	0.280	0.279	0.279	0.279	0.279	0.278	0.278	0.278	0.278
4.9	0.277	0.277	0.276	0.276	0.276	0.276	0.275	0.275	0.275	0.275
5.0	0.274	0.274	0.273	0.273	0.273	0.273	0.272	0.272	0.272	0.272
5.1	0.271	0.271	0.270	0.270	0.270	0.270	0.269	0.269	0.269	0.269
5.2	0.268	0.268	0.268	0.267	0.267	0.266	0.266	0.266	0.265	0.265
5.3	0.265	0.264	0.264	0.264	0.264	0.264	0.263	0.263	0.263	0.263
5.4	0.262	0.262	0.262	0.261	0.261	0.261	0.261	0.260	0.260	0.260
5.5	0.260	0.260	0.259	0.259	0.259	0.259	0.259	0.258	0.258	0.258
5.6	0.258	0.258	0.257	0.257	0.257	0.257	0.257	0.256	0.256	0.256
5.7	0.256	0.256	0.255	0.255	0.255	0.255	0.254	0.254	0.253	0.253
5.8	0.253	0.253	0.252	0.252	0.252	0.252	0.251	0.251	0.251	0.250
5.9	0.250	0.250	0.249	0.249	0.249	0.249	0.249	0.248	0.248	0.248
Values for V_g from 6 thru 14.9 m/s										
V_g	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
6	0.248	0.245	0.243	0.241	0.238	0.236	0.233	0.231	0.229	0.226
7	0.224	0.222	0.219	0.217	0.215	0.213	0.210	0.208	0.206	0.205
8	0.204	0.202	0.201	0.199	0.198	0.196	0.194	0.193	0.192	0.190
9	0.189	0.188	0.186	0.185	0.184	0.183	0.182	0.181	0.179	0.178
10	0.177	0.176	0.175	0.174	0.173	0.172	0.171	0.170	0.169	0.168
11	0.167	0.166	0.165	0.164	0.163	0.162	0.161	0.160	0.159	0.158
12	0.158	0.157	0.156	0.155	0.154	0.154	0.153	0.152	0.151	0.150
13	0.150	0.149	0.148	0.147	0.146	0.146	0.145	0.144	0.143	0.142
14	0.142	0.141	0.140	0.139	0.139	0.138	0.138	0.137	0.136	0.136

(continued)

Table 6 (concluded)

Values for V_g from 15 thru 30 m/s										
V_g	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
15	0.1355	0.1350	0.1336	0.1329	0.1322	0.1315	0.1309	0.1302	0.1296	0.1289
16	0.1283	0.1277	0.1271	0.1265	0.1259	0.1253	0.1247	0.1241	0.1235	0.1229
17	0.1224	0.1218	0.1213	0.1207	0.1202	0.1196	0.1191	0.1186	0.1181	0.1175
18	0.1170	0.1165	0.1160	0.1155	0.1150	0.1146	0.1141	0.1136	0.1131	0.1127
19	0.1122	0.1117	0.1113	0.1108	0.1104	0.1099	0.1095	0.1091	0.1086	0.1082
20	0.1078	0.1074	0.1070	0.1065	0.1061	0.1057	0.1053	0.1049	0.1045	0.1041
21	0.1038	0.1034	0.1030	0.1026	0.1022	0.1019	0.1015	0.1011	0.1008	0.1004
22	0.1001	0.0997	0.0994	0.0990	0.0987	0.0983	0.0980	0.0976	0.0973	0.0970
23	0.0966	0.0963	0.0960	0.0957	0.0953	0.0950	0.0947	0.0944	0.0941	0.0938
24	0.0935	0.0932	0.0929	0.0926	0.0923	0.0920	0.0917	0.0914	0.0911	0.0908
25	0.0905	0.0903	0.0900	0.0897	0.0894	0.0892	0.0889	0.0886	0.0883	0.0881
26	0.0878	0.0876	0.0873	0.0870	0.0868	0.0865	0.0863	0.0860	0.0858	0.0855
27	0.0853	0.0850	0.0848	0.0845	0.0843	0.0841	0.0838	0.0836	0.0833	0.0831
28	0.0829	0.0826	0.0824	0.0822	0.0820	0.0817	0.0815	0.0813	0.0811	0.0809
29	0.0806	0.0804	0.0802	0.0800	0.0798	0.0796	0.0794	0.0791	0.0789	0.0787
30	0.0785	--	--	--	--	--	--	--	--	--

8.3 Adverse applications

8.3.1 Operational conditions

Applications involving ambient temperatures greater than 38°C, or less than 0°C, momentary loads in excess of 300% of normal rated loads, repetitive shock loads, frequent starts and stops, external thrust loads, operation at high altitudes, worm sliding velocity in excess of 10.16 meters per second, low rotational speed (worm speed of 100 rpm or less), contaminated environments, and prime movers other than alternating current electric motors, can all have an effect on the speed reducer operation and should be referred to the manufacturer for analysis.

8.3.2 Resonant frequencies

The advent of modern measuring devices and analytical methods has made it possible to determine the resonant frequencies of the total drive system and to evaluate the effect on the speed reducer caused by operation at a speed which produces excitation at these frequencies. The resulting dynamic (vibratory) loads within the system will be superimposed on the normal running load of the speed reducer and other components of the system. Such dynamic loads can lead to destruction of the system or any of its components. In speed reducers, these vibratory loads can lead to tooth breakage, surface destruction of the worm or gear, shaft breakage or fatigue, helical pinion and gear

distress, and bearing fatigue. The manufacturer of the speed reducer is not responsible for the design of the total drive systems, and cannot be held responsible for such system effects unless special provisions are part of contract negotiations.

NOTE: This standard does not provide any allowance for the existence of resonant frequencies.

8.3.3 Overspeeds

Overspeeds contributing to external transmitted loads and dynamic loads require special analysis.

8.4 Lubrication

8.4.1 Lubrication standard

Lubricants recommended for globoidal worm gear speed reducers designed in accordance with this specification should be compounded gear oils with viscosities in accordance with ANSI/AGMA 9005-D94, *Industrial Gear Lubrication*. Compounded gear oils are petroleum based lubricants with 5 to 10 percent acidless tallow additives and are sometimes referred to as "Steam Cylinder Oils".

CAUTION: Synthetic hydrocarbons, polyglycols, and diesters are now being formulated into synthetic oils. Extreme pressure gear lubricants (EP) are petroleum based lubricants containing special chemical additives such as sulphur and phosphorous. These lubricants should not be used unless specified or approved by the speed reducer manufacturer.

8.4.2 Lubrication maintenance

It is recommended that users adhere to the reducer manufacturer's lubrication maintenance schedule. All gear oils deteriorate with age and temperature. A deteriorated or contaminated gear oil may ultimately lead to failure of the speed reducer.

8.4.3 High speed operation

Enclosed worm gear drives operating at worm sliding velocities greater than 10.16 meters per second may require special lubricants, pressurized lubrication systems, or both. Consult the gear manufacturer for recommendations.

8.4.4 Oil levels

With oil bath lubrication, oil levels must be established at a level to ensure adequate lubrication of the gearing and bearings. Required levels will vary with operating conditions, mounting position, and the configuration of slingers or oil scoops, which may be used. Most manufacturers will specify the oil level on the speed reducer itself, either by a sight glass, a designated pipe plug, or a suitable oil level indicator.

8.4.5 Temperature limitations

Maximum allowable sump temperature of an operating reducer shall not exceed 93°C. Components such as seals, shims, and gaskets may be damaged by temperatures in excess of 93°C. Where conditions result in a sump temperature in excess of 93°C, consult the manufacturer for special lubrication recommendations.

8.5 Unit thermal rating

8.5.1 Thermal capacity

Thermal capacity is defined as that input power which results in an oil sump temperature of 56°C above ambient temperature or a maximum temperature of 93°C, whichever is less. Thermal capacity is determined by establishing the allowable input power without exceeding the allowable sump temperature when tested under the following conditions.

8.5.1.1 Thermal qualifications

Qualifications which apply to the reducer thermal rating are: the reducer, when filled to specified oil level with recommended lubricant and after the unit has been "run-in", is to operate continuously at rated speed until the oil temperature in the reducer sump has stabilized.

8.5.1.2 Temperature measurement

Temperature of the lubricant is to be measured at a level slightly below the mesh of the worm and gear.

Do not measure temperature at the mesh itself. The point of temperature measurement must also be sufficiently below the static oil level to ensure proper determination of sump oil temperature.

8.5.2 Auxiliary cooling

When thermal capacity is not sufficient to maintain the indicated sump temperature, auxiliary cooling devices should be employed. Some suggested methods of auxiliary cooling are as follows:

- fan and shroud;
- cooling coils with circulating water or other coolants in the coils;
- circulating sump oil through heat exchangers or additional oil tanks.

8.6 Service factors

8.6.1 Operating conditions

When operating conditions differ from those described in 8.1.1, an appropriate service factor must be applied. See 8.6.3.

8.6.2 Equivalent power

Before an enclosed speed reducer can be designed or selected for any application, an "equivalent power" to be transmitted must be determined. In order to determine the "equivalent power", one of two methods may be used: (1) reducer input power rating is divided by the appropriate service factor, or (2) either the prime mover nameplate rating or the transmitted power is multiplied by the appropriate service factor.

The manufacturer and user must agree upon which method should dictate the selection of the gear drive. It is necessary that the gear drive selected have an input power rating in accordance with equation 27 equal to or in excess of this "equivalent power".

8.6.3 Service factor table

The listed table of service factors shown in annex D has been developed from the experience of manufacturers and users of gear drives for use in common applications and has been found to be generally satisfactory when gears are rated using AGMA standards. It is recommended that service factors for special applications be agreed upon by the user and the gear manufacturer when variations from the table may be required.

In addition to the table given in annex D, an analytical approach may be used to estimate the service factor. Such an approach is beyond the scope of this standard.

8.6.4 Operational characteristics

Some of the operational characteristics that affect the selection of the proper service factors are listed and discussed in 8.6.4.1 through 8.6.4.5.

8.6.4.1 Types of prime movers

Different types of prime movers include electric motors, steam or gas turbines, and internal combustion engines having single or multiple cylinders. Since service factors represent the normal relationship between gear design power rating and the maximum potential transmitted power, it is recommended that the service factor be applied to the nameplate power rating of the prime mover.

8.6.4.2 Starting conditions

Starting conditions where high peak loads or where frequent starts and stops occur, require special load analysis.

8.6.4.3 Systems conditions

An essential phase in the design of a system of rotating machinery is the analysis of the dynamic (vibratory) response of a system to excitation forces. It is the responsibility of the system designer to perform this dynamic analysis.

The dynamic response of a system results in additional loads imposed on the system and relative motion between adjacent elements in the system. The vibratory loads are superimposed upon the mean running load in the system and, depending upon the dynamic behavior of the system, could lead to failure of the system components.

In a gear drive these failures could occur as tooth breakage or severe surface destruction of the gear elements, shaft breakage, or bearing failure.

Any vibration analysis must consider the complete system including prime mover, gear drive, driven equipment, couplings, and foundations. The dynamic loads imposed upon a gear drive are the result of the dynamic behavior of the total system and not of the gear drive alone.

8.6.4.4 Overloads

Loads which are in excess of the nominal design loads are designated as overloads. Overload can be of momentary duration, periodic, quasi steady state or vibratory in nature. Depending on its magnitude and the number of stress cycles accumulated at

overload, it can be a fatigue or yield stress consideration.

Applications with high torque motors and motors for intermittent operations and applications where extreme repetitive shock occurs, or where high energy loads must be absorbed, as when stalling, require special consideration.

8.6.4.5 Reliability requirement and life requirement

Applications requiring a high degree of dependability or unusually long life should be given careful consideration by the user and the gear manufacturer before assigning a service factor.

8.7 Reducers used in drive systems incorporating a dynamic brake

8.7.1 Brake on gearshaft

When a dynamic brake (used to decelerate the drive system) is mounted on the slow speed shaft or gearshaft, consult the reducer manufacturer for proper reducer selection.

8.7.2 Brake on high speed or input shaft

A dynamic brake mounted in this position can create very severe dynamic reaction forces because of the wide range of backdriving efficiencies of worm gearsets. Such applications must be referred to the manufacturer for a detailed examination.

8.8 Speed reducer component design

8.8.1 Component design

For operation under loads as specified in 8.1, components must be designed and selected to meet the requirements of ANSI/AGMA 6001-D97, *Design of Components for Enclosed Gear Drives*.

8.8.2 Housings

Housings should be oil tight and of adequate design and strength for the maximum 300 percent starting loads, and must not exhibit distortions which could adversely affect gearset alignment.

8.8.3 Worm thread

The threaded portion of the worm is considered as gearing rather than shafting and is, therefore, not subject to the shafting stress limitations as specified in ANSI/AGMA 6001-D97.

8.8.4 Worm material

Worm material generally is steel, through hardened to a minimum of 32 HRC, with sufficient strength to

resist the torsional and bending stresses imposed by the maximum 300 percent permitted starting loads (see 8.1.2). Other materials may be used provided they are adequate for the application.

8.8.5 Gear material

Wormgear material generally is chill cast or centrifugally cast bronze. Generally, the bronze alloy consists of 10–12 percent tin, 0–2 percent nickel, and remainder copper. Other materials may be used provided they are adequate for the application; however, the gearset rating may be affected.

8.8.6 Oil seals

8.8.6.1 Requirements

Globoidal reducers must be furnished with oil seals on all shaft extensions. These seals should be selected to retain the lubricant in the gearbox under the normal range of applications, atmospheric conditions, and operating conditions.

8.8.6.2 Unusual operating conditions

Dusty and corrosive atmospheres, moisture or vapor laden operating conditions, high or low ambient temperatures, and other unusual operating conditions will generally require special seal compounds, seal arrangements, or other protective measures. Such operating conditions should be brought to the attention of the reducer manufacturer for recommendations.

8.8.7 Overhung load capacity

The overhung load capacity of speed reducers covered by this standard can be determined from a

consideration of both internally and externally applied loads and the limitations imposed by 8.6. The externally applied loads are to be taken in the directions which will add their effects to those of the internally applied loads. The overhung load capacity must be based on the weakest member of the reducer components. Overhung load computations must take into account starting and momentary overloads. See annex E for representative overhung load factors.

9 Gear blank design

Good wormgearing design is greatly dependent on proper gear blank design. The blank must properly support gear teeth and withstand tangential, separating and axial forces applied during tooth cutting and operation. The gear hub must serve as an effective output member, in addition to supporting the gear. This section will address effective blank design and construction. Gear materials are discussed in clause 8.

9.1 Typical gear blank configuration

Figure 12 shows typical blank designs. Gear blanks may be solid bronze, bimetal (two dissimilar metal-lurgically diffused bronzes), or constructed as composites with bronze attached to a hub typically made of cast iron, ductile iron or steel. For composite construction, adequate mechanical strength must be assured through proper lug design or selection and sizing of bolts, dowels, set screws or welds.

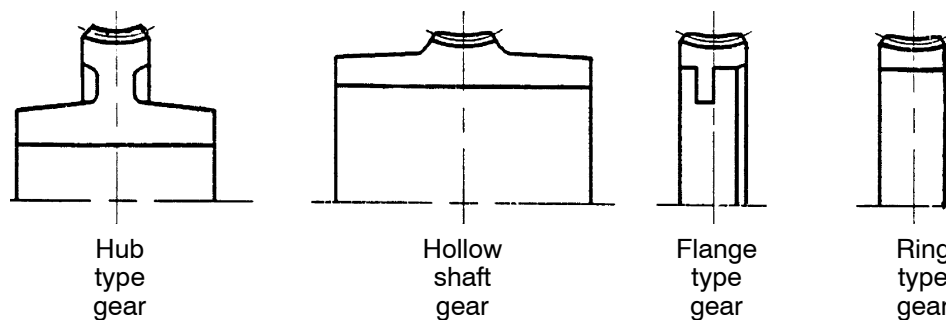


Figure 12 - Gear blank types

The amount of bronze below the root diameter is critical to properly supporting gear teeth during operation. With the exception of bimetal gears, it is recommended that the depth of material below the root diameter of coarse pitch gears equal 1.5 times the whole depth, h_2 , for general conditions, with a minimum amount of 1 times the whole depth for applications involving light, non-shock loads. For fine pitch gears, a factor greater than 1.5 times the whole depth may be necessary. When using bimetal gears, the minimum distance between the root of the gear teeth and the theoretical bond line should be 10 mm on 300 mm center distance and smaller gears, and 13 mm on gears larger than 300 mm center distance.

9.2 Gear web configuration

A web design is commonly used to reduce the amount of gear blank material. The use of a conical configuration should be considered when a gear web section is a significant portion of the blank's overall size (refer to figure 13, configuration 1). This design will add axial stiffness to the gear blanks, and it also results in reduced stresses during the casting process. For smaller or hollow output shaft gear blanks, no separate web section may exist, especially if the hub is of relatively large diameter.

9.3 Gear hub design

Good gear hub design allows for proper support of gear material and output means (solid or hollow shaft). Hub diameter and length are typically determined by the amount of material required for effective keyway strength.

Usually a single key is used for torque transfer, although multiple keys or splines are occasionally used. Cross sectional thickness, especially in the key area, must be sufficient to support operating and shock loads. Care must be taken to properly size key material in order to avoid introduction of excessive tensile stresses at keyway corners. For shaft and key stress values, consult ANSI/AGMA 6001-D97.

Typical hub design includes an interference fit between the output shaft and gear bore. The nominal interference fit is typically 0.0005 mm per mm of shaft diameter, with a maximum interference fit of 0.10 mm for cast, ductile iron or steel hubs, and 0.08 mm for bronze.

Another factor in hub design is the amount of material required over the bore to prevent excessive bore distortion due to keyway cutting. It is not uncommon to find bore distortion, in some cases equal to or exceeding total bore tolerance, after keyway cutting. This distortion does not present a problem in gearset operation, and will be eliminated during press fitting in shaft assembly.

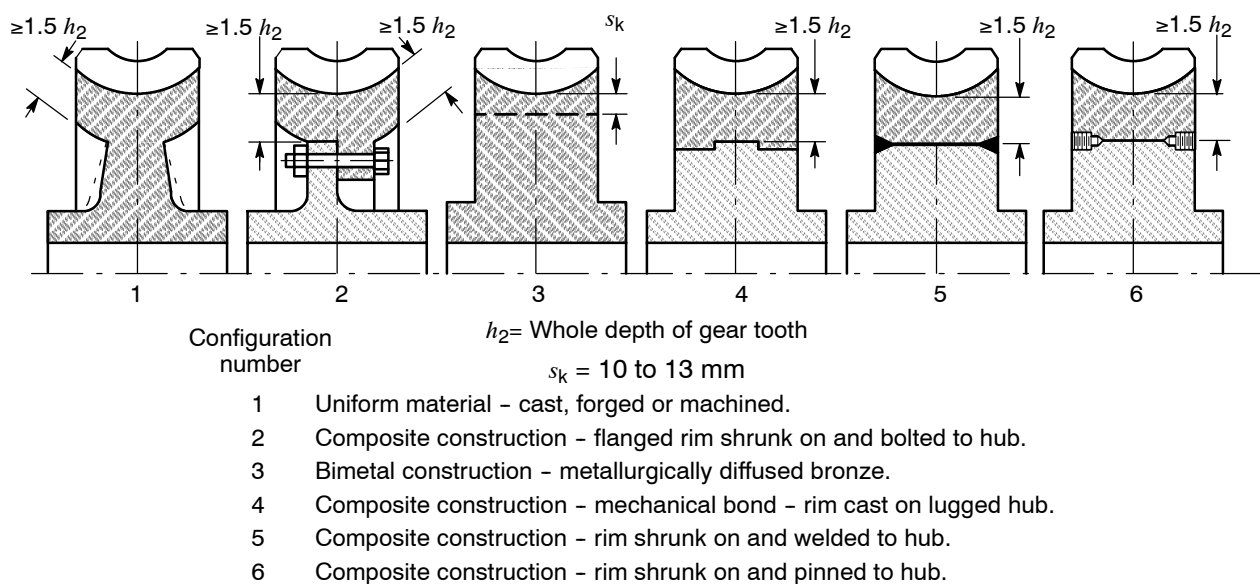


Figure 13 - Typical gear blank configuration minimum rim thickness

Annex A

(informative)

Basic assembly procedure

[The foreword, footnotes and annexes, if any, are provided for informational purposes only and should not be construed as a part of ANSI/AGMA 6135-A02, *Design, Rating and Application of Industrial Globoidal Wormgearing (Metric Edition)*.]

A.1 Purpose

The performance capability that is designed and manufactured into globoidal wormgearing can only be achieved by the correct mounting of the worm and gear in its assembly. The purpose of this annex is to provide general guidelines on proper assembly of globoidal wormgearing so as to maximize its design capabilities.

A.2 Mounting

To achieve the correct mounting, the worm and gear must be properly located in the lateral, vertical, and axial directions relative to each other. The required shaft angle must be maintained in order to provide the appropriate tooth contact pattern and backlash allowance designed into the gearing. See figure A.1.

A.3 Housing

The housing provides the worm and gear with lateral, vertical (center distance) and angular positioning. The end-covers used to maintain the output shaft

bearings also provide the axial positioning of the gear.

To achieve expected performance, the same care that goes into the design of the worm, gear and blank must be exercised in the design of the housing. The worm and gear mountings must be designed to have adequate support for all operating load conditions.

Both the worm and the gear must be restrained against lateral, vertical, axial and angular displacement. Excessive misalignment in any of these directions will reduce the load carrying capacity of the globoidal wormgear set and result in premature wear or failure.

Consideration must be given to the alignment and concentricity of opposing bearing seats and angular and vertical position relative to the mating gear element axis. Bearing carriers must be designed to maintain gear element lateral, vertical and angular positions, relative to the mating gear axis.

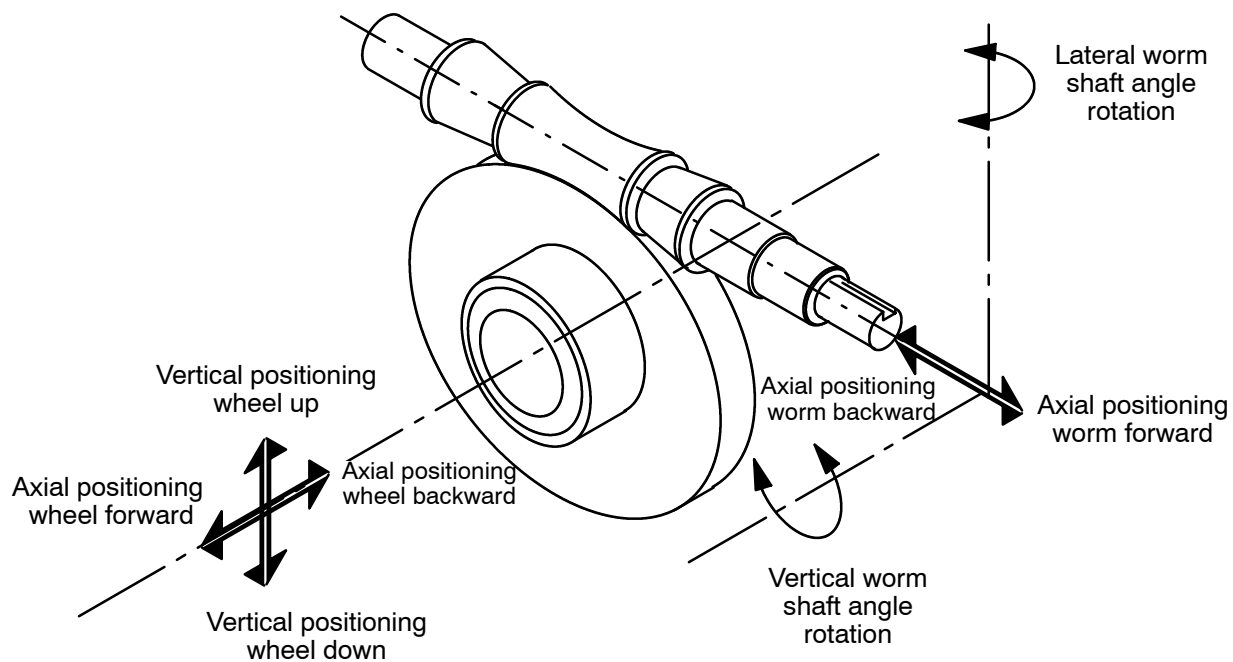


Figure A.1 - Assembly orientation

A.4 Position tolerance

Precise axial position and restraint of the worm and gear is necessary to provide the desired system backlash and contact pattern of the globoidal set. In the assembly of a globoidal worm and gear, usually the only controllable adjustment is the axial positioning of the worm and of the gear. Most designs provide a means of shimming or other method for positively locating the axial position of the worm and gear.

The gear must be axially positioned to result in it being centrally located on the worm. The worm must be axially positioned to result in it being centrally located on the gear. The final positions are established when correct gearset backlash and contact patterns are achieved.

A.5 Contact pattern check

The two basic objectives of a correct contact pattern are adequate contact area and its location, and an oil entry gap to precede the mesh. This is a visual determination. Figure A.2 provides some general information on contact patterns for full contact designs: correct patterns, incorrect patterns and means of adjustment. See clause 6 for contact patterns for Hindley designs.

A.6 Endplay check

Endplay allows for axial component thermal expansion of the worm and gear assemblies with respect to the housing, and to prevent or minimize bearing pre-load, as temperature changes from ambient to operating. Endplay must be established to achieve specific conditions at full load and operating temperature.

Assembled endplay should be kept to minimum values. Gear elements supported by tapered roller bearings will have radial play associated with endplay. Ball, spherical and cylindrical roller bearings need only be addressed from an axial gear element position standpoint. Bearing races must be fully seated in the housing and on the shaft when assembling units.

Endplay is checked by positioning a dial indicator at the end of the gear element shaft having its pointer parallel to the shaft axis. A slight axial back-and-forth force is applied to the shaft to obtain movement. For best results, the shaft should be rotated a small amount to seat the bearing rollers as readings are taken. The amount of endplay is read directly from the dial indicator.

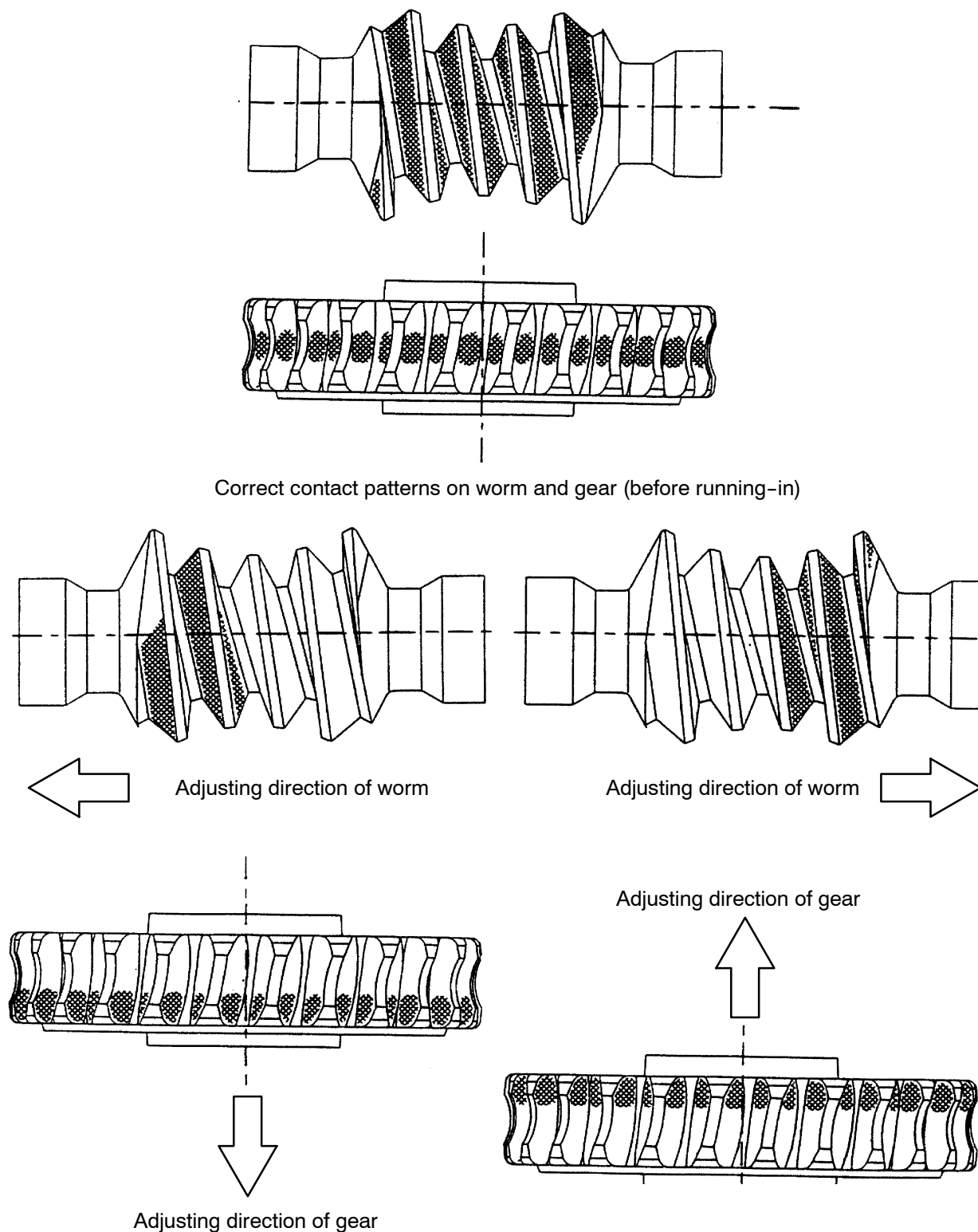


Figure A.2 - Contact patterns on worm and gear for full contact design

Annex B

(informative)

Analysis of gearing forces, worm bending stress and deflection of globoidal wormgearing

[The foreword, footnotes and annexes, if any, are provided for informational purposes only and should not be construed as a part of ANSI/AGMA 6135-A02, *Design, Rating and Application of Industrial Globoidal Wormgearing (Metric Edition)*.]

B.1 Purpose

Gearing mesh forces between the worm and gear exist which must be analyzed in order to:

- assure adequate worm root diameter design selection;
- calculate housing and bolt stresses;
- calculate bearing loading and bearing life.

The purpose of this annex is to provide a means for calculating such forces.

B.2 Tangential, separating and axial thrust forces

A globoidal gearset transmitting power is subject to reaction forces at the contact mesh as shown in figure B.1. These forces are assumed to act at the mean point of the mesh.

B.2.1 Gear tangential force

The gear tangential force component which is the worm axial thrust component is determined as follows:

$$W_{t2} = \frac{2000 T_2}{d_{m2}} = W_{a1} \quad (\text{B.1})$$

where

W_{t2} is the gear tangential force component, N;

T_2 is the gear torque, Nm;

d_{m2} is the gear mean diameter, mm;

$$d_{m2} = 2a - d_{m1};$$

W_{a1} is the worm axial thrust force component, N;

a is the center distance, mm;

d_{m1} is the worm mean diameter, mm.

$$d_{m1} = d_{a1} - h_1$$

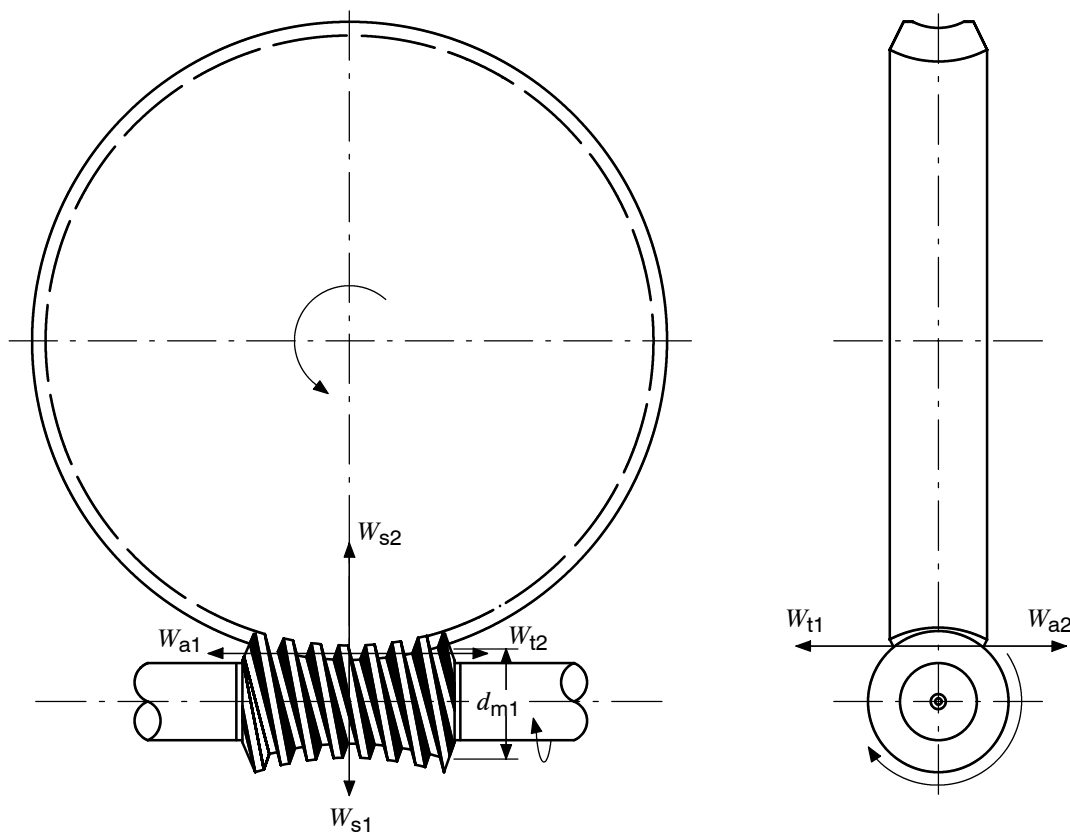


Figure B.1 - Gearing mesh forces

B.2.2 Separating force

The mesh separating force component which tends to push apart the gear and worm in mesh, and which acts parallel to the line of centers is determined as follows:

$$W_{s2} = W_{t2} \tan \alpha_x = W_{s1} \quad (\text{B.2})$$

where

W_{s2} is the gear separating force component, N;

W_{s1} is the worm separating force component, N;

W_{t2} is the gear tangential force, N

α_x is the axial pressure angle at the pitch point, degrees.

B.2.3 Gear axial thrust force

The thrust force component acting axially along the gear shaft axis is determined as follows:

$$W_{a2} = \frac{2000 T_1}{d_{m1}} = W_{t1} \quad (\text{B.3})$$

where

W_{a2} is the gear axial thrust force component, N;

T_1 is the worm torque, Nm;

$$T_1 = \frac{T_2}{u \eta}$$

W_{t1} is the worm tangential force component, N;

η is the efficiency;

u is the gear ratio.

B.3 Worm shaft bearing reactions

Referring to figure B.2, the previously calculated mesh force reactions can be used to determine the worm shaft bearing load reactions. Bearing reactions, RA and RB , are calculated as shown in equations B.4 and B.5.

$$RA = \left[\left\{ \frac{W_{a1} d_{m1}}{2(L_A + L_B)} + \frac{W_{s1} L_B}{(L_A + L_B)} \right\}^2 + \left\{ \frac{W_{t1} L_B}{(L_A + L_B)} \right\}^2 \right]^{0.5} \quad (\text{B.4})$$

$$RB = \left[\left\{ \frac{W_{a1} d_{m1}}{2(L_A + L_B)} - \frac{W_{s1} L_A}{(L_A + L_B)} \right\}^2 + \left\{ \frac{W_{t1} L_A}{(L_A + L_B)} \right\}^2 \right]^{0.5} \quad (\text{B.5})$$

where

RA is the radial reaction force of bearing A, N;

RB is the radial reaction force of bearing B, N;

L_A is the distance from the pitch point or line of centers to the effective load center of bearing A, mm;

L_B is the distance from the pitch point or line of centers to the effective load center of bearing B, mm.

NOTE: The location of the effective load center is dependent on the type of bearing used.

B.4 Worm bending stress

Once the worm shaft bearing reactions have been determined, the globoidal worm bending stress can be calculated by considering the worm as having a diameter equal to the worm root diameter at the line of centers. Refer to figure B.3 showing an equivalent beam for the worm shaft.

The worm shaft bending moment is calculated as:

$$M = \frac{RA L_A}{1000} \quad (\text{B.6})$$

$$M = \frac{RB L_B}{1000}$$

NOTE: Select the magnitude of M which is the greater of the 2 values above.

$$S_b = \frac{32\,000M}{\pi (d_{f1})^3} \quad (\text{B.7})$$

where

S_b is the worm bending stress, N/mm²;

M is the worm shaft bending moment, Nm;

d_{f1} is the worm root diameter, mm.

B.4.1 Allowable worm bending stress

The worm bending stress must be less than 17% of the ultimate tensile strength of the core material of the worm for the normal running torque of the gearset. In addition, it must be less than 75% of the yield strength of the core material of the worm for

maximum momentary overload torque which the gearset must withstand. These allowable stress levels are conservative values that reflect the usual stress concentration factors and torsional stresses present in the worm threaded portion of the worm shaft. For the definition of “momentary overload”, refer to 8.1.2.

If calculated worm bending stresses exceed the allowable worm bending stress, the worm design must be revised. Design changes include, but are not limited to: higher strength worm material, increased root diameter, reduced pressure angle, and reduced bearing span.

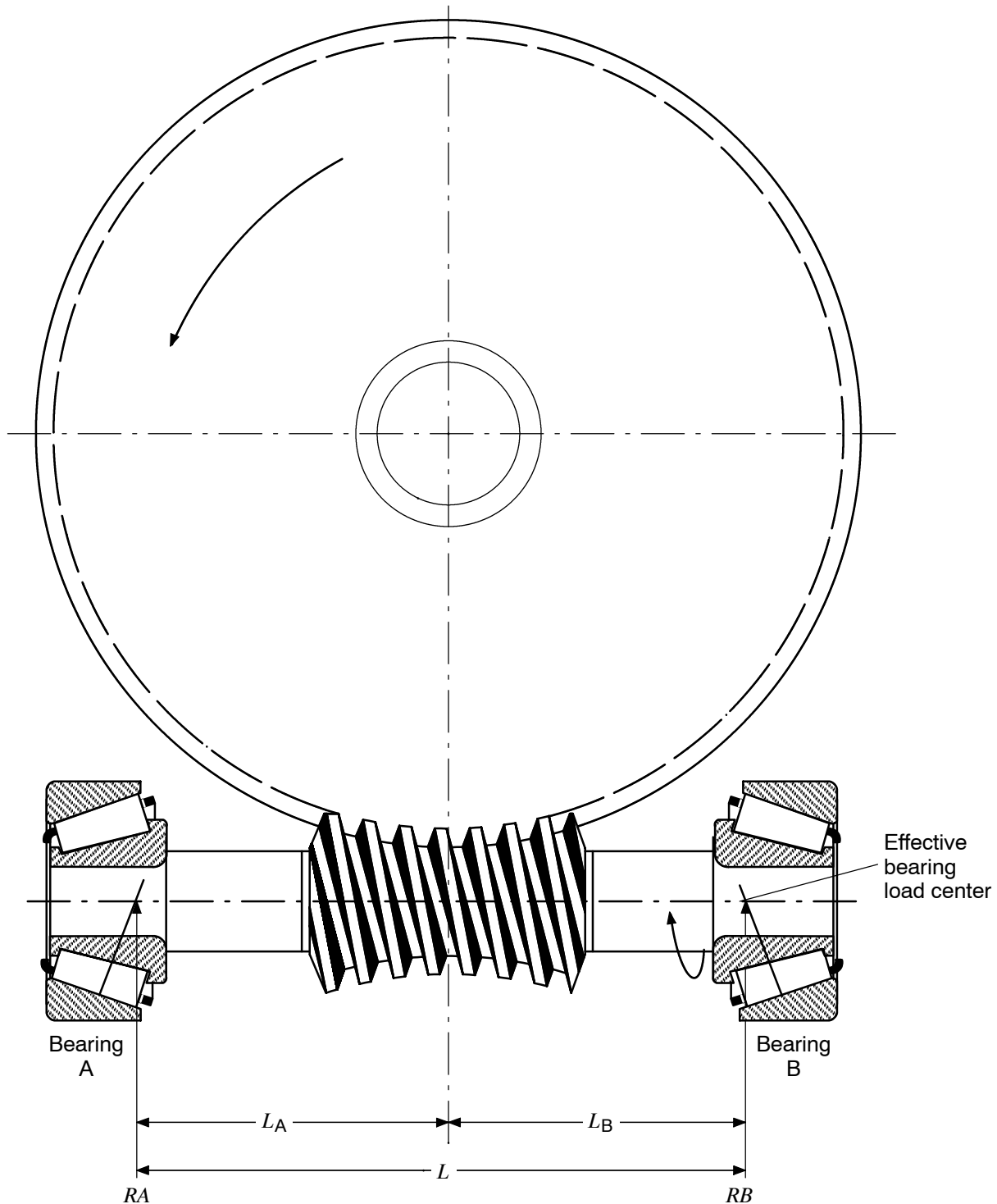


Figure B.2 - Wormshaft bearing reactions

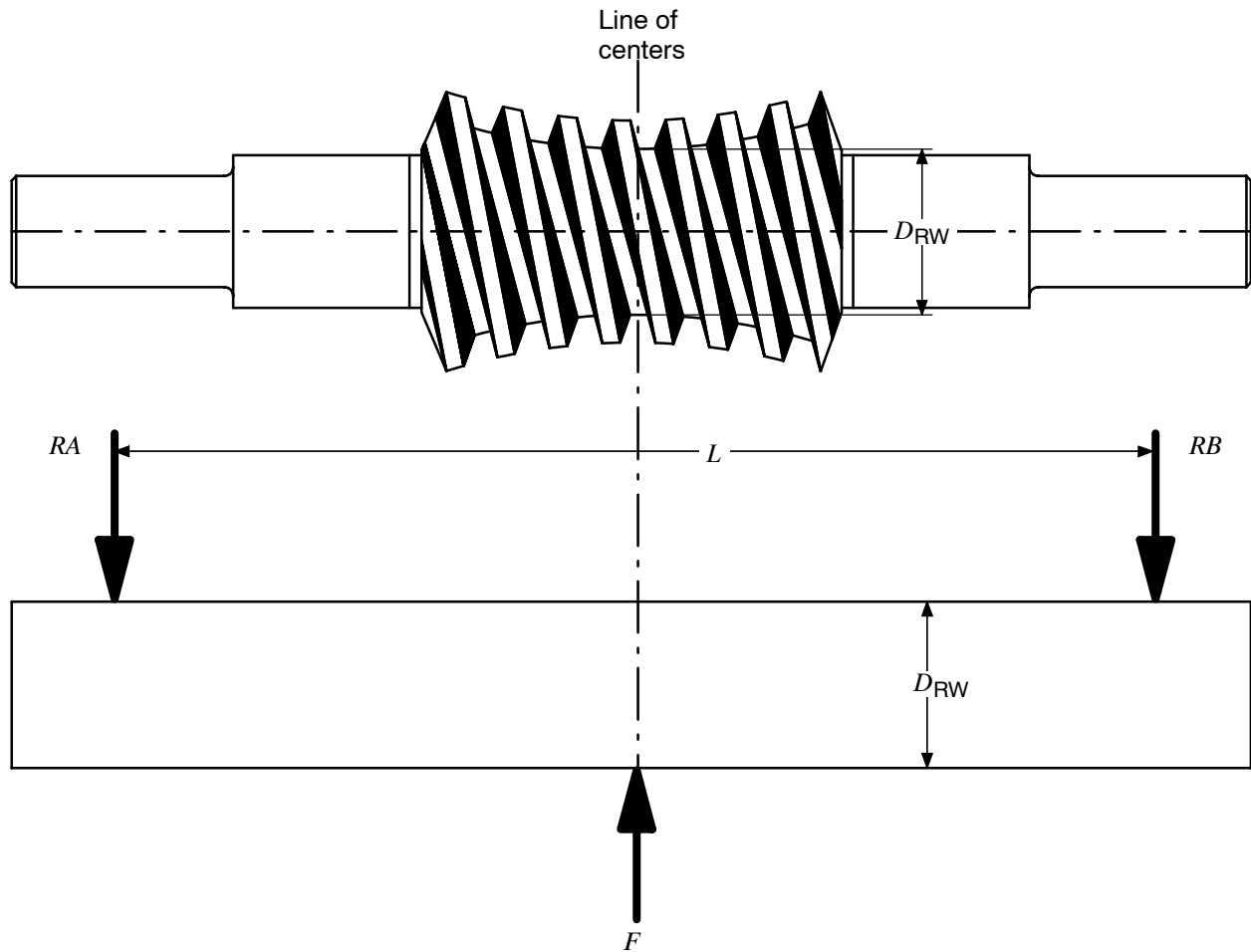


Figure B.3 - Equivalent beam for the wormshaft

B.5 Worm deflection

A conservative and approximate value of worm deflection can be calculated by using the equivalent beam method previously mentioned and shown in figure B.3 above. The procedure basically involves the calculation of the equivalentent force, F , shown in figure B.3 and based on this value, finding the worm deflection y . See equations B.8 and B.9.

$$F = RA + RB \quad (\text{B.8})$$

where

F is the equivalentent force, N.

$$y = \frac{FL^3}{3EI} (1 - k) \left\{ \frac{2}{3k} - \frac{1}{3k^2} \right\}^{\frac{3}{2}} \quad (\text{B.9})$$

where

y is the worm deflection in the central plane, mm;

L is the bearing span of the worm shaft bearings, mm;

E is the modulus of elasticity of the worm core material, N/mm² (for steel $E = 206\,850$ N/mm²);

I is the moment of inertia of the circular cross section of the worm at the root diameter, mm⁴;

k is the location factor of the equivalentent load, with $k = \text{the greater of } \frac{RA}{F} \text{ or } \frac{RB}{F}$.

$$I = \frac{\pi (d_{f1})^4}{64} \quad (\text{B.10})$$

B.5.1 Allowable worm deflection

For normal operating load conditions, the maximum allowable deflection of the worm shaft must follow the relationship shown below in order to limit the amount of worm and gear contact shift to acceptable values.

$$y \leq 0.025 \sqrt{p_{t2}}$$

where

p_{t2} is the circular pitch of the worm threads, mm.

If calculated worm deflection exceeds the allowable worm deflection based on the circular pitch shown above, a more detailed analysis is required or the worm design must be revised. Design changes include, but are not limited to increased root diameter, reduced pressure angle, and reduced bearing span.

Annex C

(informative)

Backlash measurement

[The foreword, footnotes and annexes, if any, are provided for informational purposes only and should not be construed as a part of ANSI/AGMA 6135-A01, *Design, Rating and Application of Industrial Globoidal Wormgearing (Metric Edition)*.]

Backlash allows for worm and gear runout and thermal expansion of the material. This is particularly important in globoidal gearsets because of their unique double enveloping fit, their high load carrying capacity, and sliding action.

Backlash will slightly increase from the initial value achieved during assembly to the final amount after the gear profile has developed to match the worm thread during the initial run-in period.

Backlash is commonly measured by holding the worm stationary and rotating the gear in both directions. The rotational movement of the gear is measured by a dial indicator having its pointer in the plane of rotation at or near the gear pitch circle, and parallel to a tangent to the pitch circle of the gear. The measured value is generally converted to arc minutes or degrees based on the radius of arc movement (see table C.1).

Table C.1 - Measured backlash conversion

Arc minutes	Degrees	Measurement at reference radius, mm			
		75	150	300	600
1	0.0167	0.0218	0.0436	0.0873	0.1745
2	0.0333	0.0436	0.0873	0.1745	0.3491
3	0.0500	0.0654	0.1309	0.2618	0.5236
4	0.0667	0.0873	0.1745	0.3491	0.6981
5	0.0833	0.1091	0.2182	0.4363	0.8727
6	0.1000	0.1309	0.2618	0.5236	1.0472
7	0.1167	0.1527	0.3054	0.6109	1.2217
8	0.1333	0.1745	0.3491	0.6981	1.3963
9	0.1500	0.1963	0.3927	0.7854	1.5708
10	0.1667	0.2182	0.4363	0.8727	1.7453
11	0.1833	0.2400	0.4800	0.9599	1.9199
12	0.2000	0.2618	0.5236	1.0472	2.0944
13	0.2167	0.2836	0.5672	1.1345	2.2689
14	0.2333	0.3054	0.6109	1.2217	2.4435
15	0.2500	0.3273	0.6545	1.3090	2.6180
16	0.2667	0.3491	0.6981	1.3963	2.7925
17	0.2833	0.3709	0.7418	1.4835	2.9671
18	0.3000	0.3927	0.7854	1.5708	3.1416
19	0.3167	0.4145	0.8290	1.6581	3.3162
20	0.3333	0.4363	0.8727	1.7453	3.4907
21	0.3500	0.4582	0.9163	1.8326	3.6652
22	0.3667	0.4800	0.9599	1.9199	3.8398
23	0.3833	0.5018	1.0036	2.0072	4.0143
24	0.4000	0.5236	1.0472	2.0944	4.1889
25	0.4167	0.5454	1.0908	2.1817	4.3634
26	0.4333	0.5672	1.1345	2.2690	4.5379
27	0.4500	0.5891	1.1781	2.3562	4.7125
28	0.4667	0.6109	1.2218	2.4435	4.8870
29	0.4833	0.6327	1.2654	2.5308	5.0616
30	0.5000	0.6545	1.3090	2.6181	5.2361

If the housing does not allow for direct measurement by the dial indicator on the pitch circle diameter (d_{w2}) of the teeth, a fixture can be made with an indicator located at some reference radius. This can be mounted on the free end of the output shaft with the measurement taken there.

Backlash may vary from gearset to gearset due to

manufacturing and assembly variations. Variations in tooth profiles will cause a change in backlash at different phases of tooth engagement. Worm and gear position variations that will affect backlash include vertical, lateral and worm shaft angle variations. Runout of the worm and gear assemblies may cause substantial differences between the minimum and maximum backlash at different positions.

Annex D

(informative)

Service factors

[The foreword, footnotes and annexes, if any, are provided for informational purposes only and should not be construed as a part of ANSI/AGMA 6135-A02, *Design, Rating and Application of Industrial Globoidal Wormgearing (Metric Edition)*.]

D.1 Purpose

The listed table of service factors has been developed from the experience of manufacturers and users of gear drives for use in common applications and has been found to be generally satisfactory when gears are rated using AGMA standards. It is

recommended that service factors for special applications be agreed upon by the user and the gear manufacturer when variations of the table may be required. See 8.6 for some of the operational characteristics that affect selection of proper service factors.

Table D.1 - Service factors

Hours/Day	Uniform	Moderate shock	Heavy shock	Extreme shock
1/2	1.0	1.0	1.0	1.1
1	1.0	1.0	1.0	1.2
2	1.0	1.0	1.2	1.3
10	1.0	1.2	1.3	1.5
24	1.2	1.3	1.5	1.75
NOTE: Thermal capacity must always be considered. See 8.5.				

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Annex E

(informative)

Overhung load factors

[The foreword, footnotes and annexes, if any, are provided for informational purposes only and should not be construed as a part of ANSI/AGMA 6135-A02, *Design, Rating and Application of Industrial Globoidal Wormgearing (Metric Edition)*.]

E.1 Purpose

This annex is included to provide appropriate factors for estimating purposes for speed reducers configured with overhung load.

E.2 Overhung load factor

When a chain, gear, or belt drive is mounted on the input or output shaft, the applied overhung load is calculated as a force, acting tangent to the pitch circle of the mounted member, multiplied by the applicable factor from table E.1.

Table E.1 - Overhung load factors

Drive type ¹⁾	K_{oh}
Single or multiple chain	1.00
Cut pinion run with cut gear	1.25
Synchronous belt ²⁾	1.30
Single or multiple V-belt	1.50
V-ribbed belt	1.70
Flat belt	2.50

NOTES:

1) For other drive types not listed, consult the gear drive manufacturer.

2) Synchronous belts are toothed timing belts that may have various trade names and tooth profiles.

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Annex F

(informative)

Reducer efficiency

[The foreword, footnotes and annexes, if any, are provided for informational purposes only and should not be construed as a part of ANSI/AGMA 6135-A02, *Design, Rating and Application of Industrial Globoidal Wormgearing (Metric Edition)*.]

F.1 Purpose

The efficiency-ratio-rpm curves shown below may be used when selecting reducers and motors for

specific applications, and for calculating the input power ratings of multiple reduction speed reducers.

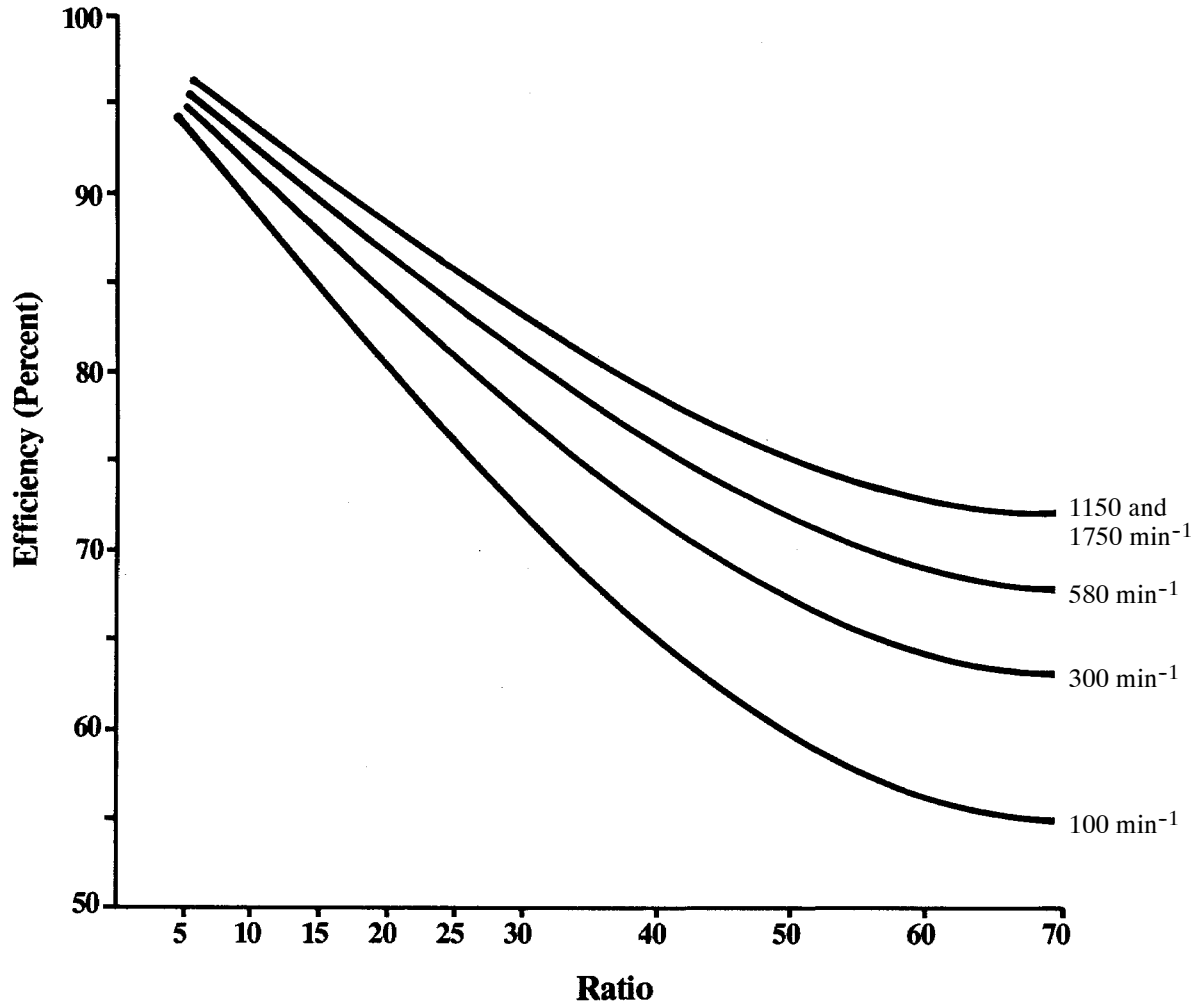


Figure F.1 - Typical efficiencies of globoidal single reduction wormgear speed reducers

NOTE: The values shown in the above curves are approximate overall reducer efficiencies which have been determined and substantiated by extensive dynamometer testing. They are for a complete reducer and include all losses within the unit from oil churning, oil seals, and bearing drag. The efficiencies shown are based on 1.0 service factor loading with the unit at normal operating temperature and with an approved lubricant. Varying conditions such as extremely cold or hot

ambient temperatures, and excessively high or low loading will effect the efficiency of the reducer.

The above curves illustrate typical efficiencies for 101.60 mm through 203.20 mm center distance reducers. Smaller sizes have efficiencies slightly less than above and larger sizes have efficiencies slightly greater than above. If precise efficiency information is needed, consult the manufacturer.

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Annex G

(informative)

Formulas for rating factors

[The foreword, footnotes and annexes, if any, are provided for informational purposes only and should not be construed as a part of ANSI/AGMA 6135-A02, *Design, Rating and Application of Industrial Globoidal Wormgearing (Metric Edition)*.]

G.1 Purpose

The purpose of this annex is to provide formulas for the rating factors given in tables 3, 4 and 5, which can be used if the rating formula is programmed for computer solution. The use of these formulas will result in values of factors Z_w , Z_u and Z_m approximately equal to those obtained from the appropriate tables.

G.2 Table 3 - Basic pressure factor, Z_w , formulas

For center distance, a , greater than or equal to 50 mm and less than or equal to 76 mm:

$$Z_w = e^{-4.338672 \left(\frac{a}{25.4} \right)^{2.7067619}} \quad (G.1)$$

For center distance, a , greater than 76 mm and less than or equal to 102 mm:

$$Z_w = 1.714961 - \frac{111.19805}{a} \quad (G.2)$$

For center distance, a , greater than 102 mm and less than 152 mm:

$$Z_w = 0.0205315 a - 1.463167 \quad (G.3)$$

For center distance, a , greater than or equal to 152 mm and less than or equal to 305 mm:

$$Z_w = e^{-4.347204 \left(\frac{a}{25.4} \right)^{2.7106719}} \quad (G.4)$$

For center distance, a , greater than 305 mm and less than 554 mm:

$$Z_w = e^{-4.380836 \left(\frac{a}{25.4} \right)^{2.72856}} \quad (G.5)$$

For center distance, a , greater than or equal to 554 mm and less than or equal to 864 mm:

$$Z_w = e^{-4.36964 \left(\frac{a}{25.4} \right)^{2.72078}} \quad (G.6)$$

For center distance, a , greater than 864 mm and less than or equal to 1000 mm:

$$Z_w = e^{-4.571009 \left(\frac{a}{25.4} \right)^{2.773764}} \quad (G.7)$$

G.3 Table 4 - Ratio correction factor, Z_u , formulas

For a ratio, u , greater than or equal to 2 and less than 4:

$$Z_u = 0.067691 + 0.105 u \quad (G.8)$$

For a ratio, u , greater than or equal to 4 and less than or equal to 6:

$$Z_u = e^{-1.431246 u^{0.51719644}} \quad (G.9)$$

For a ratio, u , greater than 6 and less than or equal to 8:

$$Z_u = \frac{1}{2.111389 - 0.075855 u} \quad (G.10)$$

For a ratio, u , greater than 8 and less than or equal to 11:

$$Z_u = e^{-0.5540175 e^{(0.0180814 u)}} \quad (G.11)$$

For a ratio, u , greater than 11 and less than or equal to 15:

$$Z_u = \frac{1}{1.246154 + \frac{2.039188}{u}} \quad (G.12)$$

For a ratio, u , greater than 15 and less than or equal to 20:

$$Z_u = e^{-0.4985559 u^{0.0646027}} \quad (G.13)$$

For a ratio, u , greater than 20 and less than or equal to 73:

$$Z_u = e^{-0.2770134 e^{\left(\frac{-0.5298417}{u} \right)}} \quad (G.14)$$

For a ratio, u , greater than 73:

$$Z_u = 0.753 \quad (G.15)$$

G.4 Table 5 - Face width and material factor, Z_m , formulas

For a center distance, a , greater than or equal to 50 mm and less than or equal to 152 mm:

$$Z_m = e^{-0.7882931 \left(\frac{a}{25.4} \right)^{0.4478914}} \quad (G.16)$$

For a center distance, a , greater than 152 mm and less than or equal to 305 mm:

$$Z_m = e^{-0.5025085 \left(\frac{a}{25.4} \right)^{0.29122965}} \quad (G.17)$$

For a center distance, a , greater than 305 mm and less than or equal to 555 mm:

$$Z_m = \frac{1}{0.894949} - 0.00030721 a \quad (G.18)$$

For a center distance, a , greater than 555 mm and less than or equal to 1000 mm:

$$Z_m = e^{0.0042213 \left(\frac{a}{25.4} \right)^{0.1046893}} \quad (G.19)$$



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